A Multi-Tissue Transcriptomic Atlas of Porcine Development Identifies Conserved Molecular Programs with Humans

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Thesis Summary

Reproducibility in pig research is hampered by the lack of a standardised, molecular definition of developmental stage. The core problem is that while the pig is an excellent preclinical model for human conditions, the molecular tempo of its development is vastly different from that of humans. Synchronising the maturation rates of the two species is essential to move preclinical research from a descriptive exercise to a predictive science. Current staging relies on chronological age or subjective morphological criteria, which fail to capture inter-individual variability and the asynchronous maturation of different organs. This thesis presents a calibrated transcriptomic atlas that systematically maps porcine development across five major tissues (brain, liver, muscle, lung, and blood), providing an objective framework for staging experimental cohorts.

By integrating large-scale RNA-seq profiles (1,924 samples) from the PigGTEx consortium with a nested cross-validation framework, this work achieves high-precision developmental staging (balanced accuracy: 0.81–0.96, mean: 0.895) and identifies tissue-resolved molecular signatures independent of chronological age. The machine learning framework employs ordinal classification using the Frank and Hall binary reduction method with Light Gradient Boosting Machine (LightGBM) classifiers and Optuna-based hyperparameter optimisation within a rigorous nested cross-validation design, enabling probabilistic stage inference that captures transitional developmental states.

To validate the biological accuracy of this atlas, a preliminary cross-species analysis was performed with a small human skeletal muscle dataset ($n = 11$), providing evidence of conservation: 97% directional concordance (64/66 genes) and significant correlation (Pearson $r = 0.69$, 95% CI: 0.54–0.80, $p = 1.09 \times 10^{-10}$) in developmental trajectories. Key regulators of neuromuscular maturation (*SORCS2*) and fibre-type specialisation (*FGFRL1*) show synchronised expression patterns between species, suggesting that the porcine atlas captures evolutionarily conserved developmental programs.

This thesis makes four key contributions: (1) the first calibrated transcriptomic atlas for porcine development; (2) a robust machine learning framework for ordinal developmental stage classification; (3) demonstration of profound tissue-specificity in developmental programs; and (4) cross-species validation revealing conserved pig-human developmental modules. This molecular staging system provides a rigorous, scalable standard for pig research, enabling precise synchronisation of experimental cohorts across studies in nutrition, growth physiology, and disease modelling.

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Contributors

Author Contributions

Tianyuan Liu conceived the study, designed the methodology, performed all computational analyses, interpreted results, and wrote the thesis and associated manuscript.
Rujing Lei contributed to manuscript editing and review.
Dr Ilyas M. Khan contributed to manuscript editing and review.
Professor Peter Theobald supervised the study, provided guidance, and edited the manuscript.

Data Contributions

The transcriptomic data used in this thesis were provided by the PigGTEx consortium (L. Chen et al., 2025; Teng et al., 2024). The human skeletal muscle data were obtained from the study (Schaiter et al., 2024), available through the NCBI Gene Expression Omnibus (GEO: GSE257558).

Publications Arising from this Thesis

Journal Articles

1. **Liu, T.**, Lei, R., Khan, I. M., & Theobald, P. (2026). A Multi-Tissue Transcriptomic Atlas of Porcine Development Identifies Conserved Molecular Programs with Humans. *[Manuscript in preparation for submission]*.

Data and Code Availability

All code required to reproduce the findings of this thesis are available at: https://github.com/TianYuan-Liu/pig-dev-stage

The PigGTEx data used in this study are available at: https://piggtex.farmgtex.org/

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Chapter 1

General Introduction

1.1 The Domestic Pig as a Research Model

The domestic pig (*Sus scrofa domesticus*) occupies a unique position at the intersection of agricultural science and biomedical research, serving as both a major food production species and an increasingly important model organism for human biology and disease (Lunney et al., 2021; Meurens et al., 2012; Swindle et al., 2012). This dual role has driven substantial investment in understanding porcine biology at the molecular level, yet fundamental challenges remain in standardising the characterisation of developmental state across research studies.

1.1.1 Historical Importance in Biomedical Research

The use of pigs in biomedical research has a long and distinguished history, dating back to the pioneering surgical studies of the nineteenth century. Early investigators recognised that the pig's anatomical and physiological similarities to humans made it an invaluable platform for developing surgical techniques and testing therapeutic interventions (Swindle et al., 2012). Throughout the twentieth century, pigs became established models for cardiovascular research, with the porcine heart serving as a testing ground for prosthetic valves, coronary artery bypass procedures, and cardiac catheterisation techniques that would later be translated to human medicine (Swindle et al., 2012).

The completion of the pig genome sequence in 2012 marked a watershed moment in porcine research, providing unprecedented molecular resources for comparative studies with humans (Groenen et al., 2012). This genomic foundation has enabled researchers to move beyond purely anatomical comparisons to interrogate the molecular basis of organ development, disease susceptibility, and drug metabolism. The subsequent development of transcriptomic atlases, including the PigGTEx resource (L. Chen et al., 2025; Teng et al., 2024), has further expanded the scope of molecular investigations in this species.

Today, pigs are employed across a remarkable breadth of research applications. In regenerative medicine, porcine organs are being developed for xenotransplantation, with recent clinical trials demonstrating the feasibility of pig-to-human heart transplantation (Lunney et al., 2021). In pharmaceutical development, minipigs serve as non-rodent species for toxicological testing, providing safety data that complements rodent studies (Swindle et al., 2012). In nutritional science, pigs are used to model human dietary interventions and gastrointestinal function (Pluske et al., 2018). Across all these applications, the ability to accurately characterise the developmental state of experimental animals is essential for ensuring reproducibility and enabling meaningful comparisons across studies.

1.1.2 Anatomical and Physiological Similarities to Humans

The pig's value as a biomedical model derives fundamentally from its remarkable anatomical and physiological similarities to humans. These similarities extend across multiple organ systems and developmental processes, making the pig a more representative model for human biology than traditional rodent species in many contexts (Lunney et al., 2021).

The cardiovascular system exemplifies this similarity. The porcine heart is comparable to the human heart in size, structure, and coronary artery anatomy, making it an ideal model for studying cardiac development, ischaemic heart disease, and interventional cardiology (Swindle et al., 2012). The pig's haemodynamic parameters, including blood pressure, heart rate, and cardiac output, fall within ranges similar to those of humans, facilitating the translation of cardiovascular interventions (Galbas et al., 2024; Hannon et al., 1990; Lelovas et al., 2014).

The gastrointestinal tract represents another area of strong correspondence. Pigs and humans share similar digestive physiology, gut microbiome composition, and nutrient absorption mechanisms (Pluske et al., 2018). This makes the pig particularly valuable for nutritional studies and research into gastrointestinal diseases. The porcine immune system also shares important features with humans, including similar organisation of secondary lymphoid tissues and comparable responses to pathogens (Mair et al., 2014).

From a developmental perspective, pigs share several important features with humans. Pigs are polytocous (giving birth to multiple offspring), and both species experience relatively prolonged postnatal development compared to rodents and undergo sexual maturation at comparable biological ages relative to lifespan (Ahn et al., 2014). These developmental parallels make the pig an appropriate model for studying growth, maturation, and the transitions between life stages.

1.1.3 Advantages Over Rodent Models

While rodents remain the workhorses of biomedical research due to their small size, short generation time, and well-characterised genetics, pigs offer distinct advantages for certain research questions that justify their higher maintenance costs and longer experimental timelines.

Size is perhaps the most obvious advantage. Adult pigs can reach body masses of 100–300 kg, similar to adult humans, while even minipig breeds typically weigh 20–40 kg (Simianer & Köhn, 2010; Swindle et al., 2012). This size similarity enables the testing of medical devices, surgical procedures, and drug delivery systems at scales directly relevant to human application. Organ dimensions, tissue volumes, and blood flow rates in pigs more closely approximate human values than those of mice or rats.

Lifespan represents another important consideration. Pigs can live for up to 27 years under optimal conditions, with domestic production animals typically reaching market weight at 5–6 months (Hoffman & Valencak, 2020; Swindle et al., 2012). This intermediate lifespan, combined with a developmental timeline that spans months rather than weeks, allows researchers to study postnatal development and maturation processes at temporal resolutions impossible to achieve in short-lived rodent models.

Metabolically, pigs share important features with humans that differ from rodents. Drug metabolism pathways, particularly those involving cytochrome P450 enzymes, show greater similarity between pigs and humans than between rodents and humans (Swindle et al., 2012). This metabolic correspondence improves the predictive validity of pharmacokinetic and toxicological studies conducted in pigs.

The pig genome, at approximately 2.8 billion base pairs distributed across 18 autosomes plus sex chromosomes (Groenen et al., 2012), is similar in size and organisation to the human genome. Comparative analyses have revealed extensive synteny (conservation of gene order) between pig and human chromosomes, facilitating the translation of genetic findings between species (Groenen et al., 2012; Warr et al., 2020). This genomic similarity extends to the transcriptomic level, where gene expression patterns in porcine tissues often parallel those observed in corresponding human tissues (Cardoso-Moreira et al., 2019).

1.1.4 Agricultural Significance and Research Infrastructure

Beyond its biomedical applications, the pig's economic importance as a food animal has driven substantial investment in research infrastructure that benefits basic science investigations. The global pork industry produces over 100 million tonnes of meat annually (Kim et al., 2023), creating powerful incentives to understand and optimise porcine growth, development, and health.

This agricultural context has generated extensive resources for pig research. Large-scale breeding programmes have produced well-characterised genetic lines with defined pedigrees extending back many generations. Consortium efforts such as PigGTEx have assembled transcriptomic datasets spanning dozens of tissues and hundreds of individuals (Teng et al., 2024). Industry investments in genomic selection have produced dense genotyping arrays and imputation reference panels that enable genome-wide association studies at scale (Groenen et al., 2012; Ramos et al., 2009).

The dual agricultural and biomedical utility of pig research creates synergies that benefit both communities. Fundamental discoveries about porcine biology often have direct applications for improving production efficiency, while the economic resources generated by the pork industry support research infrastructure that advances biomedical applications. This mutually beneficial relationship has accelerated progress in understanding porcine molecular biology over the past two decades.

1.2 The Challenge of Developmental Staging

Despite the extensive molecular resources now available for pig research, a critical gap undermines reproducibility across studies (Baker, 2016): the lack of a standardised, molecular definition of developmental stage. This deficiency compromises experimental design, limits cross-study comparisons, and introduces confounding variability that reduces statistical power and interpretive clarity.

1.2.1 Current Methods: Chronological Age Versus Biological Maturity

Contemporary pig research typically defines developmental state using one of two approaches: chronological age or morphological staging criteria. Neither approach adequately captures the biological complexity of postnatal development (Boeyer et al., 2020), and both introduce sources of variability that compromise experimental reproducibility.

Chronological age, measured in days from birth, represents the simplest and most commonly used staging criterion. Researchers specify the age of experimental animals at the time of sampling or intervention, enabling straightforward comparisons across studies. However, chronological age is a poor proxy for biological maturity because it fails to account for the substantial variation in developmental trajectories among individuals (Boeyer et al., 2020).

The limitations of chronological staging become apparent when considering the diversity of factors that influence developmental rate. Genetic background, nutrition, environmental conditions, and health status all affect the pace of maturation (Boeyer et al., 2020), leading to situations where animals of identical chronological age may differ substantially in their biological state. A piglet from a large litter with limited access to maternal nutrition may develop more slowly than a singleton from a well-nourished sow, yet both animals would be assigned the same developmental stage based solely on age (Hole et al., 2018; Yuan et al., 2015).

Morphological staging offers an alternative approach that attempts to capture biological state more directly. Researchers may classify animals based on physical characteristics such as body weight, tooth eruption, testicular descent, or secondary sexual characteristics. These morphological markers provide some information about developmental progress, but they suffer from subjective assessment, inter-observer variability, and limited precision (Boeyer et al., 2020).

Moreover, morphological staging typically captures only whole-organism development, whereas different organ systems may mature at different rates (Cardoso-Moreira et al., 2019). This asynchrony creates situations where an animal might appear developmentally advanced by one criterion but immature by another. For studies targeting specific tissues or physiological processes, morphological staging of the whole organism may provide limited information about the developmental state of the tissues of interest.

1.2.2 Limitations of Morphological Staging Criteria

The historical reliance on morphological staging in developmental biology reflects the practical constraints faced by researchers before molecular tools became widely available. Visual assessment of physical characteristics required no specialised equipment and could be performed rapidly on large numbers of animals. However, the advent of high-throughput molecular profiling has revealed the inadequacy of these traditional approaches (Teschendorff & Horvath, 2025).

Morphological development proceeds in a non-linear fashion characterised by periods of rapid change interspersed with plateaus of relative stability (Boeyer et al., 2020). This non-linearity makes morphological markers unreliable indicators of developmental state during transitional periods. An animal undergoing rapid morphological change may be difficult to classify unambiguously, while two animals at slightly different developmental stages may appear morphologically identical during a plateau phase.

The subjectivity inherent in morphological assessment introduces additional variability. Different observers may apply staging criteria differently, even when following standardised protocols (Boeyer et al., 2020). This inter-observer variability reduces the reproducibility of morphological staging and limits the precision of cross-study comparisons. Training and calibration can reduce but not eliminate these subjective elements.

Perhaps most fundamentally, morphological staging assumes that external physical characteristics accurately reflect internal physiological state. This assumption breaks down when considering the development of organs such as the brain, liver, or immune system, where functional maturation may proceed independently of external morphological changes. A piglet that appears physically mature may harbour immature neural circuits or an underdeveloped immune repertoire (Pluske et al., 2018), with profound implications for studies targeting these systems.

1.2.3 Inter-Individual Variability and Asynchronous Organ Development

One of the most significant challenges in developmental staging arises from the recognition that different organs and tissues mature at different rates within the same individual. This asynchrony means that no single staging criterion can accurately capture the developmental state of all tissues simultaneously.

Evidence for asynchronous development comes from multiple sources. Longitudinal studies of skeletal maturation in humans have demonstrated that bone age can deviate substantially from chronological age, with some individuals showing accelerated maturation while others are delayed (Boeyer et al., 2020). Similar variability has been documented in the development of the immune system, central nervous system, and reproductive organs (Georgountzou & Papadopoulos, 2017; Semple et al., 2013).

In pigs, asynchronous development creates particular challenges for studies of growth physiology. The gastrointestinal tract undergoes rapid maturation during the weaning transition, with changes in enzyme expression, absorptive capacity, and barrier function occurring over days to weeks (Pluske et al., 2018). However, muscle tissue continues to develop and remodel for months after weaning (Cardoso-Moreira et al., 2019), while the brain may not reach full maturity until well into adulthood (Semple et al., 2013). A staging system based on gastrointestinal markers would provide limited information about muscle or brain development in the same animals.

Littermate studies have revealed the extent of developmental variability even among genetically related individuals reared under identical conditions. Piglets from the same litter may differ substantially in birth weight, growth rate, and developmental trajectory, reflecting stochastic elements of development as well as intra-uterine positioning and access to maternal resources (Hole et al., 2018; Yuan et al., 2015).

These littermate differences demonstrate that controlling for genetics and environment is insufficient to standardise developmental state.

1.2.4 Impact on Experimental Reproducibility

The consequences of inadequate developmental staging extend throughout the research process, from experimental design to data interpretation and cross-study comparison. Variability introduced by developmental differences inflates experimental noise, reduces statistical power, and can lead to both false-positive and false-negative findings (Baker, 2016).

Consider a hypothetical study investigating the effects of a dietary intervention on muscle gene expression in growing pigs. If experimental animals vary in developmental stage at the time of sampling, this variation will contribute to the variance in gene expression measurements, potentially obscuring treatment effects. The study may require larger sample sizes to achieve adequate statistical power, increasing costs and animal usage. Alternatively, if the study is conducted with inadequate sample sizes, genuine treatment effects may go undetected.

The problem is compounded when attempting to compare results across studies conducted at different institutions using different staging criteria. One research group may define "weaning" as the point when piglets are separated from the sow, while another may use a weight threshold or a specific chronological age. These operational differences make it difficult to determine whether discrepant findings reflect genuine biological variation or merely differences in the developmental composition of experimental groups.

For pig research to achieve its potential as a translational platform (Lunney et al., 2021; Swindle et al., 2012), standardisation of developmental staging is essential. Researchers must be able to specify the developmental state of their experimental animals with precision and reproducibility, enabling meaningful comparisons across studies and the accumulation of knowledge over time.

1.3 Molecular Approaches to Biological Age

The limitations of morphological and chronological staging have motivated the development of molecular approaches that estimate biological age directly from tissue measurements (Duan et al., 2022; Hannum et al., 2013; Horvath, 2013). These molecular clocks leverage the systematic changes in epigenetic marks, gene expression patterns, or other molecular features that accompany the ageing process, providing quantitative estimates of biological state that can complement or replace traditional staging methods (BI Ageing Clock Team et al., 2017; Muralidharan et al., 2025; E. D. Sun et al., 2025).

1.3.1 Epigenetic Clocks

The discovery that DNA methylation patterns change systematically with age has enabled the development of “epigenetic clocks” that predict chronological age with remarkable accuracy (BI Ageing Clock Team et al., 2017; Duan et al., 2022; Horvath, 2013). These clocks are constructed by training machine learning models on methylation data from individuals of known age, identifying the CpG sites whose methylation levels are most strongly associated with chronological age, and combining these sites into a predictive score.

The seminal epigenetic clock developed by Horvath demonstrated that chronological age could be predicted with a median absolute error of approximately 3.6 years across diverse tissue types and individuals ranging from prenatal development to advanced age (Horvath, 2013). Subsequent refinements have improved accuracy and revealed that the deviation between epigenetic age and chronological age—termed “age acceleration”—predicts mortality risk and disease outcomes independently of chronological age (Teschendorff & Horvath, 2025).

Epigenetic clocks have been developed for multiple species, including pigs (Schachtschneider et al., 2021). These porcine epigenetic clocks have demonstrated that DNA methylation-based age prediction is feasible in this species and have been used to investigate the effects of diet, obesity, and other factors on biological ageing in pig models. However, the requirement for DNA methylation profiling, typically using array-based or sequencing-based methods, limits the practical application of epigenetic clocks in settings where transcriptomic data are more readily available.

A universal pan-mammalian epigenetic clock has now been established across 185 species (Lu et al., 2023), demonstrating that biological age prediction can be translated directly between humans and other mammals. Although tissue-specific methylation clocks can be constructed, they primarily capture the gradual, cumulative drift of CpG methylation with age rather than the abrupt transcriptional switches that mark developmental phase transitions (Hoshino et al., 2019). Epigenetic age can therefore align two species globally while organ-level maturation remains asynchronous. Transcriptomic profiling complements methylation-based clocks by providing a direct, functional readout of cellular state, capturing phase-specific gene programs that define tissue maturation.

Recent methodological advances have expanded the repertoire of epigenetic clocks, with new versions designed to predict specific outcomes such as mortality risk, functional capacity, or the pace of ageing (Teschendorff & Horvath, 2025). Furthermore, the field is moving towards higher resolution with the development of spatial and cell-type-specific aging clocks (Muralidharan et al., 2025; E. D. Sun et al., 2025), which disentangle the contributions of distinct cell populations to the overall tissue aging signal. These second- and third-generation clocks demonstrate that molecular age estimation can capture aspects of biological state beyond simple chronological age, opening possibilities for more nuanced characterisation of developmental and ageing processes.

1.3.2 Transcriptomic Signatures and Gene Expression Clocks

While epigenetic clocks have received substantial attention, gene expression patterns offer an alternative molecular window into developmental state. Transcriptomic profiles capture the functional state of tissues at a given moment (Z. Wang et al., 2009), reflecting the integrated output of regulatory networks that control cellular identity, function, and response to environmental stimuli.

The Genotype-Tissue Expression (GTEx) project established a foundational resource for understanding transcriptomic variation across human tissues (The GTEx Consortium, 2020). By profiling gene expression in dozens of tissues from hundreds of individuals, GTEx has enabled the identification of tissue-specific expression patterns, the characterisation of genetic influences on gene expression, and the development of reference atlases for human tissue states. More recently, the developmental GTEx (dGTEx) project has extended this approach to capture transcriptomic changes across postnatal development in humans and non-human primates (Coorens et al., 2025).

Transcriptomic approaches to biological age estimation exploit the systematic changes in gene expression that accompany development and ageing. During postnatal development, tissues undergo characteristic transcriptional programs that reflect cellular differentiation, functional maturation, and adaptation to postnatal challenges (Cardoso-Moreira et al., 2019). By identifying the genes whose expression changes most consistently with age, researchers can construct transcriptomic clocks analogous to epigenetic clocks.

The advantages of transcriptomic approaches include the wide availability of RNA-seq data, the functional interpretability of gene expression changes, and the potential to capture tissue-specific developmental trajectories. Unlike DNA methylation, which provides an integrated record of epigenetic state (Teschendorff & Horvath, 2025), gene expression captures the immediate functional state of cells, potentially providing more sensitive detection of transitional states and environmental influences.

1.3.3 Multi-Tissue Versus Single-Tissue Approaches

A critical consideration in developing molecular staging systems is whether to target a single tissue or attempt to capture development across multiple tissues simultaneously. Both approaches have advantages and limitations that must be weighed against the specific requirements of particular research applications (Duan et al., 2022).

Single-tissue approaches focus on one tissue type, developing markers that capture the developmental trajectory of that specific tissue. This approach can achieve high accuracy for the targeted tissue but provides no information about the development of other organs (Hannum et al., 2013). For research questions focused on a single organ system, such as studies of muscle development or liver metabolism, single-tissue staging may be appropriate and efficient.

Multi-tissue approaches attempt to identify molecular markers that track development across diverse tissue types (Horvath, 2013). The appeal of this approach is the potential to capture whole-organism developmental state with a single measurement. However, the asynchrony of development across tissues complicates this goal, as markers that are informative in one tissue may be uninformative or even misleading in another (Meer et al., 2018).

A middle ground involves developing tissue-specific markers within a unified framework that allows comparison across tissues (BI Ageing Clock Team et al., 2017). This approach acknowledges that different tissues follow distinct developmental trajectories while enabling researchers to characterise the developmental state of multiple tissues within the same experimental animals. The resulting tissue-resolved staging system provides more complete information about organismal development than any single-tissue approach.

1.3.4 Current State of Molecular Staging in Pigs

Despite the development of epigenetic clocks for pigs and the accumulation of extensive transcriptomic resources through projects like PigGTEx, a comprehensive molecular staging system for porcine development has been lacking. Existing porcine epigenetic clocks focus primarily on ageing rather than developmental staging (Schachtschneider et al., 2021), and no transcriptomic framework has been established for inferring developmental stage from gene expression data.

The PigGTEx resource provides an unprecedented foundation for developing such a framework (L. Chen et al., 2025; Teng et al., 2024). This consortium has assembled RNA-seq profiles from dozens of tissues across hundreds of pigs spanning the postnatal lifespan, creating a comprehensive reference for porcine gene expression across tissues and ages. However, translating this resource into a practical staging system requires analytical approaches that can identify developmental markers, train predictive models, and validate the biological relevance of the resulting staging framework.

The gap between available molecular resources and practical staging tools motivates the present thesis. By developing and validating a transcriptomic framework for porcine developmental staging, this work aims to provide the pig research community with a rigorous, reproducible alternative to traditional staging methods.

1.4 Research Gap and Objectives

1.4.1 Identification of the Research Gap

The preceding sections have established the importance of the pig as a biomedical and agricultural model, highlighted the limitations of current developmental staging approaches, and reviewed molecular methods that offer potential solutions. From this foundation, a clear research gap emerges: the absence of a validated, tissue-resolved transcriptomic framework for inferring developmental stage in pigs.

This gap has concrete consequences for the pig research community. Without standardised molecular staging, researchers cannot precisely specify the developmental state of experimental animals, limiting reproducibility and cross-study comparison (Baker, 2016). The asynchronous development of different tissues means that whole-organism staging methods fail to capture the complexity of postnatal development, potentially introducing uncontrolled variability into studies targeting specific organ systems.

Furthermore, the relationship between porcine and human development at the molecular level remains incompletely characterised. While anatomical and physiological similarities between species are well documented, the conservation of developmental gene expression programs has not been systematically investigated. Understanding this conservation is essential for establishing the pig's validity as a translational model for human developmental biology.

1.4.2 Research Aims and Objectives

This thesis addresses the identified research gap through four interconnected objectives:

Objective 1: To construct a comprehensive transcriptomic atlas of porcine development across major tissue types, integrating publicly available RNA-seq data from the PigGTEx consortium with calibrated developmental stage annotations.

Objective 2: To develop and validate a machine learning framework capable of inferring developmental stage directly from tissue-specific gene expression profiles, providing probabilistic estimates that capture transitional states and quantify classification uncertainty.

Objective 3: To characterise the molecular signatures underlying developmental staging in each tissue, identifying the genes and pathways that drive developmental transitions and comparing these signatures across tissues to assess the tissue-specificity of developmental programs.

Objective 4: To validate the biological accuracy of the porcine developmental atlas through cross-species comparison with human transcriptomic data, testing whether conserved molecular programs underlie postnatal development in both species.

Together, these objectives provide a systematic approach to developing and validating molecular staging tools for pig research. The resulting framework will enable researchers to characterise the developmental state of experimental animals with unprecedented precision, supporting improved experimental design and enhanced reproducibility across the field.

1.4.3 Scope and Limitations

The scope of this thesis is defined by the available data resources and the practical requirements of a functional staging system. The analysis focuses on five major tissues (brain, liver, muscle, lung, and blood) selected for their biological importance and the availability of sufficient sample sizes for machine learning analysis. Other tissues profiled in the PigGTEx resource but lacking adequate developmental stage representation are excluded from the primary analysis.

The cross-species validation is limited to skeletal muscle tissue due to the availability of matched developmental transcriptomic data in humans. While this limitation constrains the generalisability of cross-species findings, muscle tissue represents an important target for developmental research and provides a rigorous test of the conservation hypothesis.

The machine learning framework developed in this thesis is designed for research applications where tissue samples can be processed for RNA-seq analysis. The approach is not intended for field applications requiring rapid, point-of-care staging decisions. Future work may extend the framework to more accessible assay platforms, but such extensions are beyond the scope of the present thesis.

1.5 Thesis Structure

This thesis is organised into six chapters, proceeding from background and methods through results and interpretation to conclusions and future directions.

Chapter 1: General Introduction (the present chapter) has established the context for this research, reviewing the pig's role as a biomedical model, the challenges of developmental staging, and molecular approaches to biological age estimation. The chapter concludes with the identification of the research gap and the statement of research objectives.

Chapter 2: Literature Review provides a comprehensive review of the scientific literature relevant to this thesis. Topics include developmental biology and classical staging systems, transcriptomic technologies and their application to developmental research, machine learning methods for biological classification, and cross-species comparative approaches. The chapter synthesises this literature to identify the methodological approaches most appropriate for addressing the research objectives.

Chapter 3: Materials and Methods describes the data sources, analytical methods, and computational approaches employed in this thesis. The chapter provides detailed documentation of the machine learning framework, including data preprocessing, model architecture, hyperparameter selection, and evaluation metrics. Sufficient detail is provided to enable reproduction of the analyses.

Chapter 4: Results presents the findings of the thesis in four main sections: (1) characterisation of the transcriptomic atlas and sample distribution; (2) evaluation of model performance across tissues; (3) analysis of developmental gene signatures and their tissue-specificity; and (4) cross-species validation with human skeletal muscle. Results are presented with appropriate statistical analyses and visualisations.

Chapter 5: Discussion interprets the results in the context of the broader literature, addressing the biological implications of the findings, the methodological contributions of the work, and the practical applications for pig research. The chapter provides a critical evaluation of limitations and identifies directions for future research.

Chapter 6: Conclusions summarises the key contributions of the thesis, highlights the implications for the field, and provides closing remarks on the significance of molecular staging for improving reproducibility in pig research.

Chapter 2

Literature Review

This chapter provides a comprehensive review of the scientific literature relevant to the development of a transcriptomic framework for porcine developmental staging. The review synthesises knowledge from developmental biology, transcriptomics, machine learning, and comparative genomics, critically evaluating current methodologies and identifying the approaches most suitable for addressing the research objectives outlined in Chapter 1.

2.1 Developmental Biology: From Morphology to Molecules

Developmental biology has undergone a profound transformation over the past century, evolving from a discipline focused on morphological description to one centred on molecular mechanisms. This section reviews the history of developmental staging systems, examines the specific features of porcine postnatal development, and evaluates the transition from morphological to molecular approaches for characterising developmental state.

2.1.1 Classical Developmental Staging Systems

The systematic classification of developmental stages has a history extending back more than a century. Early embryologists recognised the need for standardised reference points that would enable meaningful comparison of observations across different laboratories and species. The resulting staging systems provided a common vocabulary for describing development and facilitated the accumulation of knowledge about embryonic and postnatal processes.

The Carnegie staging system, developed in the early twentieth century for human embryonic development, exemplifies the classical approach to developmental staging (O'Rahilly & Müller, 2010). This system defines 23 stages spanning fertilisation through the eighth week of gestation, with each stage characterised by specific morphological features. The Carnegie stages have proven remarkably durable, remaining in widespread use more than a century after their initial formulation.

For mouse embryonic development, the Theiler stages serve a similar function, defining 26 stages from fertilisation through birth (Theiler, 1989). These stages are defined by external morphological landmarks such as limb bud formation, eye development, and digit separation. The Theiler staging system has become the standard reference for mouse developmental biology, enabling precise specification of embryonic age in this important model organism.

The extension of staging systems to postnatal development presents additional challenges (Boeyer et al., 2020). Unlike embryonic development, which proceeds through a relatively conserved sequence of morphological transformations, postnatal development involves the gradual maturation of already-formed organs and tissues. The landmarks that define embryonic stages—limb bud emergence, neural tube closure, eye formation—have no direct analogues in postnatal development. Researchers have therefore relied on chronological age, body weight, or tissue-specific functional milestones to characterise postnatal developmental state.

2.1.2 Porcine Developmental Milestones

Postnatal development in pigs proceeds through a series of physiological transitions that are fundamental to understanding the biology of this species (Swindle et al., 2012). These transitions provide the biological rationale for the developmental staging system employed in this thesis.

The **neonatal period** (0–20 days) represents the initial phase of postnatal life, during which piglets are entirely dependent on maternal nutrition and care. During this period, the gastrointestinal tract is adapted to digest milk, with high expression of lactase and limited capacity to handle solid food (Lallès et al., 2007). The immune system relies heavily on passive immunity transferred from the sow through colostrum, with the piglet's own adaptive immune responses still immature (Summerfield et al., 2015). Thermoregulation is poorly developed, requiring external heat sources or huddling behaviour to maintain body temperature (Le Dividich & Sève, 2000).

The **early childhood** period (21–59 days) encompasses the weaning transition, one of the most significant physiological challenges in the porcine lifespan. Weaning typically occurs at 21–28 days in commercial production systems (Pluske et al., 2018), though it may be later in other contexts. This transition requires rapid adaptation of the gastrointestinal tract to handle solid food, including changes in digestive enzyme expression, gut morphology, and barrier function (Pluske et al., 2018). The weaning transition is associated with temporary growth depression and increased susceptibility to enteric pathogens (Lallès et al., 2007; Pluske et al., 1997), reflecting the physiological stress of this developmental milestone.

The **pre-pubertal period** (60–149 days) represents a phase of rapid growth and tissue deposition. During this period, pigs achieve the highest rates of body weight gain, with muscle and adipose tissue accumulating rapidly (Whittemore & Kyriazakis, 2006). Skeletal development continues, with bones increasing in both length and density (Reiland, 1978). The immune system matures, with the development of a diverse repertoire of antigen-specific responses (Bæk et al., 2019). Neural development continues, with refinement of synaptic connections and maturation of cognitive functions (Semple et al., 2013).

The **post-pubertal period** (150–365 days) encompasses sexual maturation and the approach to adult physiological function. In gilts (female pigs), puberty typically occurs at 5–7 months of age (Swindle et al., 2012), marked by the onset of oestrous cycling. In boars (male pigs), puberty is characterised by the development of mature spermatogenesis and the expression of sexual behaviour (Lents et al., 2020). Beyond reproductive maturation, this period involves continued refinement of tissue function and the establishment of adult metabolic homeostasis (Whittemore & Kyriazakis, 2006).

The **adult period** (>365 days) represents the mature physiological state, with fully developed organ systems and reproductive capacity. While growth slows substantially after the first year of life, tissue remodelling continues throughout adulthood in response to physiological demands, reproductive status, and environmental conditions (Swindle et al., 2012).

These developmental periods provide a biologically meaningful framework for classifying porcine developmental state. However, the boundaries between periods are not sharp, and individuals may transition between stages at different rates depending on genetic background, nutrition, and environmental factors.

2.1.3 Tissue-Specific Developmental Trajectories

A critical insight from developmental biology is that different tissues follow distinct developmental trajectories, maturing at different rates and through different mechanisms. This asynchrony complicates any attempt to define whole-organism developmental state from a single measurement.

The nervous system exemplifies prolonged postnatal development. Brain growth continues for months after birth, with ongoing neurogenesis, synaptic pruning, and myelination (Jiang & Nardelli, 2016; Kang et al., 2011). The development of cognitive functions follows a protracted timeline, with learning and memory capabilities continuing to mature well into adolescence. Molecular studies have revealed waves of gene expression changes accompanying these developmental processes, with distinct transcriptional programs active at different stages of neural maturation (Kang et al., 2011).

Skeletal muscle follows a different developmental trajectory. Muscle fibres are established prenatally, but postnatal development involves substantial changes in fibre type composition, contractile protein expression, and metabolic capacity (Lefaucheur et al., 1995). The transition from fast glycolytic to slow oxidative fibres, the development of neuromuscular junctions, and the establishment of adult contractile function all occur during postnatal life (Lefaucheur, 2010). These changes are reflected in transcriptomic profiles that shift systematically with age.

The liver undergoes rapid functional maturation during the neonatal and weaning periods. The transition from fetal to adult hepatic function involves changes in metabolic pathways, drug-metabolising enzyme expression, and bile acid synthesis (Gordillo et al., 2015; Si-Tayeb et al., 2010). These changes enable the liver to handle the metabolic demands of postnatal life, including the processing of nutrients from solid food and the detoxification of xenobiotics.

The immune system represents perhaps the most dramatic example of postnatal development. While some immune functions are present at birth, the development of a mature adaptive immune repertoire requires years of exposure to antigens and the expansion of antigen-specific lymphocyte populations (Georgountzou & Papadopoulos, 2017). The timing and extent of immune maturation are strongly influenced by environmental factors, including pathogen exposure and microbial colonisation of the gut.

This tissue-specific asynchrony has important implications for developmental staging. A staging system based on liver function would not accurately reflect brain development in the same animal, and vice versa. The recognition of this asynchrony motivates the tissue-resolved approach adopted in this thesis, which develops separate staging models for each tissue while enabling comparison across tissues within a unified framework.

2.1.4 Critical Evaluation of Morphological Staging Limitations

The transition from morphological to molecular staging approaches reflects a growing recognition of the limitations of traditional methods. While morphological staging has served developmental biology well for over a century, several fundamental limitations constrain its utility for modern research applications.

Morphological staging relies on the assumption that external features accurately reflect internal developmental state. This assumption holds reasonably well during embryonic development, when morphological transformations are tightly coupled to underlying molecular and cellular changes (O'Rahilly & Müller, 2010). However, the assumption becomes progressively less reliable during postnatal development, when external appearance may remain relatively stable while internal tissues continue to mature.

The subjectivity of morphological assessment introduces variability that is difficult to quantify or control (Reiland, 1978). Different observers may interpret staging criteria differently, and the same observer may apply criteria inconsistently across time or contexts. While standardised training and calibration can reduce this variability, it cannot be eliminated entirely.

Perhaps most fundamentally, morphological staging provides limited resolution for characterising developmental state. The number of discrete stages that can be reliably distinguished based on morphological criteria is inherently limited by the precision of visual assessment (Reiland, 1978). Molecular approaches, by contrast, provide continuous measurements that can capture subtle variations in developmental state and enable the detection of transitional states between discrete stages.

These limitations motivate the development of molecular staging approaches that can provide more objective, reproducible, and precise characterisation of developmental state.

2.2 Transcriptomics in Developmental Research

The development of high-throughput technologies for measuring gene expression has transformed the study of development, enabling researchers to characterise the transcriptional state of cells and tissues with unprecedented comprehensiveness. This section reviews the evolution of transcriptomic technologies, evaluates their application to developmental research, and examines the tissue-specific expression atlases that provide the foundation for molecular staging approaches.

2.2.1 RNA Sequencing Technology Evolution and Capabilities

RNA sequencing (RNA-seq) has become the dominant technology for transcriptome profiling, superseding earlier microarray-based approaches (Marioni et al., 2008). The technology involves converting RNA molecules to complementary DNA, fragmenting the cDNA, sequencing the fragments, and mapping the resulting reads to a reference genome or transcriptome. The number of reads mapping to each gene provides a measure of that gene's expression level.

The first RNA-seq studies, published in the late 2000s (Mortazavi et al., 2008; Z. Wang et al., 2009), demonstrated the technology's advantages over microarrays: broader dynamic range, greater sensitivity for lowly expressed genes, and the ability to detect novel transcripts and splice variants. Over the subsequent decade, improvements in sequencing chemistry, library preparation methods, and computational analysis have increased both the throughput and accuracy of RNA-seq experiments.

Current RNA-seq protocols can profile the expression of tens of thousands of genes simultaneously, with sufficient sensitivity to detect transcripts present at a few copies per cell (Z. Wang et al., 2009). The declining cost of sequencing has enabled large-scale studies profiling hundreds or thousands of samples, generating datasets of unprecedented size and comprehensiveness.

For developmental research, RNA-seq provides the ability to characterise the transcriptional programs that drive tissue maturation (Cardoso-Moreira et al., 2019). By comparing expression profiles across developmental stages, researchers can identify the genes whose expression changes most dramatically during development, infer the regulatory networks that control these changes, and relate transcriptional state to physiological function.

2.2.2 Bulk Versus Single-Cell Transcriptomics for Developmental Studies

A fundamental distinction in transcriptomic methodology is between bulk and single-cell approaches. Bulk RNA-seq measures the average expression across all cells in a tissue sample, while single-cell RNA-seq (scRNA-seq) measures expression in individual cells. Both approaches have advantages and limitations for developmental studies.

Bulk RNA-seq provides robust, cost-effective profiling of tissue-level expression. The averaging across cells means that bulk profiles are relatively reproducible and can be compared meaningfully across samples and studies (X. Li & Wang, 2021). However, bulk profiling obscures cellular heterogeneity, conflating expression changes in different cell types into a single measurement. During development, when the cellular composition of tissues may be changing, bulk profiles can be difficult to interpret.

Single-cell RNA-seq addresses the heterogeneity problem by profiling individual cells (Schaum et al., 2018), enabling the identification of cell types, the characterisation of cell-type-specific expression, and the reconstruction of developmental trajectories at cellular resolution. Single-cell studies have revealed previously unrecognised cellular diversity in developing tissues and have identified rare cell populations that would be undetectable in bulk profiles.

However, scRNA-seq has its own limitations. Current technologies capture only a fraction of the transcripts in each cell, resulting in sparse, noisy data (X. Li & Wang, 2021; Zheng et al., 2023). The technical variability in single-cell data is substantially higher than in bulk data, requiring sophisticated computational methods to separate biological signal from technical noise. The cost per sample remains higher than for bulk sequencing, limiting the number of samples that can be profiled.

For the purposes of developmental staging, bulk RNA-seq offers practical advantages. The PigGTEx resource, which provides the foundation for this thesis, employs bulk RNA-seq, providing a large dataset with comprehensive coverage of tissues and developmental stages. While single-cell approaches may offer advantages for understanding the cellular basis of developmental changes, bulk profiles are well-suited for developing robust staging classifiers that can be applied to new samples.

2.2.3 Tissue-Specific Expression Atlases

The generation of comprehensive expression atlases has been a major focus of transcriptomic research, providing reference datasets that enable researchers to understand tissue-specific expression patterns and compare their samples to established baselines.

The Genotype-Tissue Expression (GTEx) project represents the most ambitious human transcriptomic atlas to date (The GTEx Consortium, 2020). GTEx has profiled gene expression in 54 tissue types from nearly 1,000 deceased donors, creating a comprehensive reference for human tissue-specific expression. The resource has enabled the characterisation of expression quantitative trait loci (eQTLs), the identification of tissue-specific regulatory elements, and the development of methods for predicting gene expression from genotype.

While GTEx focuses primarily on adult tissues, the developmental GTEx (dGTEx) project aims to extend this resource to capture developmental changes (Coorens et al., 2025). dGTEx will profile tissues from birth through adulthood in humans, as well as developmentally matched non-human primates, providing an unprecedented resource for understanding transcriptomic changes during postnatal development.

For pigs, the PigGTEx resource provides a comparable atlas of tissue-specific expression (L. Chen et al., 2025; Teng et al., 2024). PigGTEx has profiled gene expression across dozens of tissues from hundreds of pigs, including animals at various developmental stages. Complementing this, recent studies have further characterised the porcine RNA editome across diverse tissues, revealing the landscape of post-transcriptional modifications (Long et al., 2024). These resources provide the foundation for the developmental staging framework developed in this thesis.

2.2.4 Gene Expression as a Proxy for Biological Age

The systematic changes in gene expression that accompany development and ageing provide the biological basis for transcriptomic staging approaches. These changes reflect the coordinated regulation of thousands of genes as tissues mature, adapt to physiological challenges, and eventually senesce.

During development, gene expression changes are driven by developmental regulatory networks that control cell differentiation, tissue morphogenesis, and functional maturation (Cardoso-Moreira et al., 2019). These networks involve transcription factors that activate or repress large sets of target genes, signalling pathways that transmit information between cells, and epigenetic mechanisms that establish stable patterns of gene expression.

The genes most strongly associated with developmental stage are often those encoding proteins with direct roles in tissue function. In muscle, developmental changes in the expression of contractile proteins, metabolic enzymes, and ion channels reflect the maturation of contractile and metabolic function (Lefaucheur, 2010). In the brain, changes in expression of neurotransmitter receptors, synaptic proteins, and myelination-related genes reflect the development of neural circuits (Kang et al., 2011).

By measuring the expression of developmentally regulated genes, researchers can infer the developmental state of a tissue without relying on morphological assessment or chronological age. This approach forms the basis of transcriptomic staging systems, which use machine learning methods to combine information from many genes into a robust prediction of developmental state.

2.2.5 Normalisation Methods: TPM Versus FPKM Versus Counts

A critical technical consideration in transcriptomic analysis is the choice of normalisation method. Different normalisation approaches produce expression values with different properties, and the choice of method can affect downstream analyses.

Raw read counts represent the most direct measure of gene expression from RNA-seq data but are affected by both library size (total sequencing depth) and gene length. Genes with more reads will appear more highly expressed, as will longer genes that produce more fragments. Comparisons across samples or genes require normalisation to account for these factors.

Fragments Per Kilobase of transcript per Million mapped reads (FPKM) normalises for both library size and gene length (Mortazavi et al., 2008), producing values that can be compared across genes within a sample and across samples. However, FPKM values can be distorted by highly expressed genes that consume a large fraction of sequencing reads, and the length normalisation may not be appropriate for all applications.

Transcripts Per Million (TPM) provides an alternative normalisation that addresses some limitations of FPKM (Zhao et al., 2021). TPM first normalises for gene length, then normalises for library size, ensuring that the sum of TPM values within a sample is constant. This makes TPM values more comparable across samples than FPKM values.

For the analyses in this thesis, TPM values are used as provided by the PigGTEx resource. TPM normalisation is appropriate for developmental staging applications because it enables meaningful comparison of expression levels across samples from different individuals and developmental stages.

2.2.6 Critical Evaluation of Transcriptomic Approaches

While transcriptomic approaches offer substantial advantages over morphological staging, they also have limitations that must be acknowledged.

Bulk RNA-seq averages expression across all cells in a tissue sample, potentially obscuring important heterogeneity. During development, changes in cellular composition may contribute to apparent changes in bulk expression, complicating the interpretation of developmental signatures. A gene that appears to increase in expression during development might actually be expressed at constant levels in a specific cell type that becomes more abundant.

Technical variability in RNA-seq data can introduce noise that obscures biological signal. Batch effects, arising from differences in sample processing, library preparation, or sequencing, can dominate biological variation if not properly controlled. The PigGTEx data used in this thesis were generated using standardised protocols to minimise batch effects, but some residual technical variation is inevitable.

The choice of TPM normalisation, while appropriate for many applications, has its own limitations. TPM values are relative measures that sum to a constant within each sample, meaning that changes in the expression of highly expressed genes can affect the apparent expression of all other genes. For developmental staging, this relative nature of TPM is acceptable because the staging models are trained and applied within the same normalisation framework.

Despite these limitations, transcriptomic approaches provide substantial advantages for developmental staging. The ability to measure thousands of genes simultaneously enables the identification of robust developmental signatures that are not dependent on any single gene. The quantitative nature of RNA-seq data enables the development of predictive models that provide continuous estimates of developmental state rather than discrete classifications.

2.3 Machine Learning in Biomedical Research

Machine learning has emerged as an essential tool for analysing the high-dimensional data generated by modern transcriptomic technologies. This section reviews the application of machine learning to biomedical classification problems, with particular emphasis on the methods and considerations relevant to developmental staging.

2.3.1 Overview of Machine Learning in Biology

Machine learning encompasses a diverse set of computational methods that learn patterns from data and use these patterns to make predictions or decisions (Greener et al., 2022). In biomedical research, machine learning has found applications in disease diagnosis, drug discovery, image analysis, and many other domains.

Supervised learning, the paradigm most relevant to developmental staging, involves training a model on labelled examples to predict labels for new, unlabelled data. In the context of this thesis, the training data consist of gene expression profiles from samples of known developmental stage, and the goal is to predict developmental stage for new samples.

The choice of machine learning algorithm depends on the characteristics of the data and the requirements of the application. For gene expression data, which typically involve thousands of features (genes) measured in relatively few samples, methods that handle high-dimensional data and incorporate regularisation to prevent overfitting are particularly appropriate (Greener et al., 2022).

2.3.2 Classification Versus Regression for Age Prediction

A fundamental choice in developing a molecular staging system is whether to treat developmental stage as a categorical (classification) or continuous (regression) variable. Both approaches have been used in previous work on molecular age prediction (Frank & Hall, 2001; Horvath, 2013), each with distinct advantages.

Classification approaches divide development into discrete stages and train models to assign samples to one of these stages. This approach aligns with the traditional practice of defining developmental periods and is appropriate when the boundaries between stages have biological meaning. Classification also enables straightforward interpretation: a sample is assigned to a specific stage with an associated confidence.

Regression approaches treat age as a continuous variable and train models to predict chronological age directly. This approach captures the continuous nature of development and can provide more precise estimates of biological age. However, regression may be less appropriate when the relationship between gene expression and age is non-linear or when the goal is to identify discrete developmental transitions.

The ordinal nature of developmental stages—the fact that stages have a natural ordering from young to old—suggests a third approach: ordinal classification. Ordinal classification methods exploit the ordering of stages while maintaining the categorical framework, enabling models to learn that adjacent stages are more similar than distant stages.

2.3.3 Ordinal Classification: Theory and Methods

Ordinal classification addresses problems where the classes have a natural ordering but the distances between classes are not necessarily equal. Developmental staging is a prototypical ordinal problem: an infant is more similar to early childhood than to adult, but the difference between stages is not quantified on a continuous scale.

Standard classification methods ignore the ordering of classes, treating all misclassifications as equally severe. For ordinal problems, this is suboptimal: misclassifying an infant as early childhood is a less serious error than misclassifying an infant as adult. Ordinal methods incorporate this ordering into the learning process, encouraging models to make errors that respect the ordinal structure.

The Frank and Hall binary reduction method provides an elegant approach to ordinal classification (Frank & Hall, 2001). Given K ordered classes, the method decomposes the ordinal problem into $K-1$ binary classification problems. The kth binary classifier predicts whether the true class is greater than k. The class probabilities are then computed from the binary predictions:

$$P(y = c) = P(y > c - 1) - P(y > c) \tag{2.1}$$

where $P(y > -1) = 1$ and $P(y > K - 1) = 0$.

This decomposition has several advantages. It transforms the ordinal problem into a series of standard binary problems, enabling the use of any binary classifier. The resulting class probabilities respect the ordinal structure by construction. The method is straightforward to implement and has proven effective across a range of ordinal classification tasks (Frank & Hall, 2001).

Alternative approaches to ordinal classification include threshold models, which fit a single regression function with multiple thresholds separating classes; CORAL (consistent rank logits), which modifies the neural network loss function to encourage ordinal consistency (Cao et al., 2020); and ordinal variants of specific classifiers such as ordinal random forests (Hornung, 2020).

2.3.4 Ensemble Methods: Random Forest Versus Gradient Boosting

Ensemble methods combine multiple weak learners into a strong learner, typically achieving better predictive performance than any single model. For gene expression data, two families of ensemble methods have proven particularly effective: random forests and gradient boosting.

Random forests construct an ensemble of decision trees (Breiman, 2001), each trained on a bootstrap sample of the data with a random subset of features considered at each split. The final prediction is obtained by averaging (for regression) or voting (for classification) across trees. Random forests are robust to overfitting, handle high-dimensional data well, and provide measures of feature importance.

Gradient boosting constructs an ensemble sequentially, with each new tree trained to correct the errors of the previous trees. This iterative refinement can produce more accurate predictions than random forests but requires careful tuning of hyperparameters to prevent overfitting. Modern implementations such as XGBoost (T. Chen & Guestrin, 2016) and LightGBM have made gradient boosting highly efficient and widely accessible.

For gene expression classification, gradient boosting often achieves superior accuracy compared to random forests, particularly when combined with appropriate regularisation (Ke et al., 2017). The ability of gradient boosting to model complex interactions between features makes it well-suited for transcriptomic data, where developmental signatures may involve coordinated changes across many genes.

2.3.5 LightGBM: Architecture and Advantages

Light Gradient Boosting Machine (LightGBM) is a gradient boosting framework developed by Microsoft that has become widely used in machine learning competitions and applications (Ke et al., 2017). LightGBM introduces several innovations that improve both computational efficiency and predictive accuracy.

Gradient-based One-Side Sampling (GOSS) reduces the computational cost of finding optimal splits by focusing on instances with large gradients, which contribute most to the loss function. Exclusive Feature Bundling (EFB) groups mutually exclusive features together, reducing the effective dimensionality of the data without loss of information (Ke et al., 2017).

For transcriptomic data, LightGBM's efficiency enables rapid model training even with thousands of features, while its regularisation capabilities help prevent overfitting on the relatively small sample sizes typical of biological datasets. The framework provides built-in support for feature importance calculation, enabling identification of the genes most informative for classification.

LightGBM's leaf-wise tree growth strategy, which grows trees by splitting the leaf with maximum gain rather than level-by-level, can produce more accurate models than level-wise approaches, though it requires careful control of tree depth to prevent overfitting.

2.3.6 Feature Selection Strategies

Gene expression data typically involve many more features (genes) than samples, creating challenges for machine learning algorithms (Greener et al., 2022). Feature selection methods address this challenge by identifying the most informative features and excluding uninformative or redundant features.

Filter methods rank features based on statistical measures of their association with the target variable, selecting the top-ranked features for model training. For classification problems, univariate measures such as the F-statistic or mutual information can identify features that differ significantly across classes. Filter methods are computationally efficient and can be applied as a preprocessing step before model training.

Embedded methods incorporate feature selection into the model training process. Tree-based methods such as random forests and gradient boosting naturally provide measures of feature importance based on the contribution of each feature to prediction accuracy. Features with high importance can be retained while those with low importance are excluded.

Wrapper methods evaluate subsets of features based on the performance of a model trained on that subset. While wrapper methods can identify optimal feature combinations, they are computationally expensive and may overfit the feature selection to the training data.

For the developmental staging framework in this thesis, feature selection is performed using the embedded importance measures from LightGBM. This approach leverages the model's ability to identify informative features while avoiding the computational cost of exhaustive subset evaluation.

2.3.7 Evaluation Metrics for Ordinal Problems

Evaluation of ordinal classification models requires metrics that account for the ordered nature of the classes (Greener et al., 2022). Standard classification metrics such as accuracy treat all errors as equivalent, failing to distinguish between minor and major misclassifications.

Mean Absolute Error (MAE) measures the average distance between predicted and true class indices. For ordinal problems, MAE penalises predictions that are far from the true class more heavily than predictions that are close, making it appropriate for evaluating ordinal consistency.

Spearman's rank correlation coefficient (ρ) measures the monotonic relationship between predicted and true classes. A high Spearman correlation indicates that the model correctly orders samples by developmental stage, even if the exact stage classifications are not always correct.

Cohen's Kappa with quadratic weighting provides another ordinal-aware metric, weighting disagreements by the squared distance between classes. Quadratic Kappa penalises distant misclassifications more heavily than adjacent misclassifications, aligning with the ordinal structure of developmental stages.

Balanced accuracy addresses class imbalance by averaging recall across classes (Chicco & Jurman, 2020), ensuring that performance on rare classes contributes equally to the overall metric. For developmental data, where some stages may be underrepresented, balanced accuracy provides a fairer assessment of model performance than simple accuracy.

2.3.8 Critical Evaluation of Machine Learning Approaches in Transcriptomics

While machine learning provides powerful tools for analysing transcriptomic data, several considerations must be addressed to ensure reliable and interpretable results.

Overfitting is a constant concern when the number of features exceeds the number of samples. Machine learning models can learn spurious patterns that happen to be present in the training data but do not generalise to new samples. Proper validation procedures, including held-out test sets and cross-validation, are essential for assessing true model performance.

Feature selection can introduce bias if not performed carefully. If feature selection is based on information from the test set, performance estimates will be overly optimistic. The appropriate practice is to perform feature selection using only training data and to evaluate performance on truly independent test data.

Interpretability of machine learning models varies considerably. While tree-based methods provide feature importance measures, these measures may not correspond to true biological importance. High importance may indicate that a feature is useful for classification without indicating a causal role in development.

Despite these caveats, machine learning approaches provide substantial advantages for developmental staging. The ability to combine information from thousands of genes into a single prediction enables robust staging that is not dependent on any single biomarker. The probabilistic framework of ordinal classification provides not just point predictions but measures of uncertainty that can inform downstream analyses.

2.4 Cross-Species Comparative Biology

A central question for the pig as a biomedical model is whether its developmental processes reflect those of humans. Cross-species comparative biology provides the tools to address this question, testing whether molecular programs are conserved across evolutionary time.

2.4.1 Evolutionary Conservation of Developmental Programs

Comparative studies have revealed remarkable conservation of developmental programs across diverse animal species (Cardoso-Moreira et al., 2019). The genes that control body plan formation, organ development, and cellular differentiation are often shared across species that diverged hundreds of millions of years ago.

This conservation reflects the ancient origins of developmental regulatory networks and the evolutionary constraints that maintain their function. Core developmental genes, such as the Hox genes that control body segment identity, are present in nearly all animals and perform similar functions across species. Signalling pathways such as Wnt, Notch, and Hedgehog are conserved from flies to humans, controlling similar developmental processes in distantly related organisms (Qin et al., 2024; Zong et al., 2009).

The conservation of developmental programs at the molecular level provides the foundation for using animal models to understand human development. If the genes and pathways controlling development are shared between species, insights gained from model organisms can inform understanding of human biology.

However, conservation is not absolute. Species differ in developmental timing, in the relative sizes and functions of organs, and in adaptations to their specific ecological niches. Understanding which aspects of development are conserved and which diverge is essential for interpreting cross-species comparisons.

2.4.2 Pig-Human Comparisons in Developmental Biology

The pig occupies a useful position in the evolutionary tree for comparative studies with humans. As mammals, pigs and humans share the fundamental features of mammalian development: internal fertilisation, placental gestation, lactation, and extended postnatal care. However, the approximately 80 million years since the divergence of pigs and humans (Kumar et al., 2022) have allowed substantial divergence in developmental timing and tissue-specific adaptations.

Comparative genomic studies have revealed extensive synteny between pig and human chromosomes, with large blocks of genes maintained in the same order in both species (Groenen et al., 2012). This syntenic conservation facilitates the identification of orthologous genes and the comparison of expression patterns between species.

At the transcriptomic level, studies have compared gene expression across tissues in pigs and humans, revealing both conserved and divergent patterns (Cardoso-Moreira et al., 2019; Gui et al., 2024). Core tissue-specific expression signatures are largely conserved, with liver-expressed genes in pigs showing enrichment for the same pathways as liver-expressed genes in humans. However, the precise expression levels and the regulatory elements controlling expression may differ between species.

Developmental comparisons between pigs and humans have been more limited, partly due to the historical scarcity of developmental transcriptomic data in both species. The availability of resources such as PigGTEx and dGTEx is now enabling more systematic comparison of developmental trajectories.

2.4.3 Methodological Considerations for Cross-Species Analysis

Cross-species comparison of gene expression requires careful attention to methodological issues that can confound interpretation.

Ortholog identification is the first challenge. While most mammalian genes have clear orthologs in other mammals, some genes have undergone lineage-specific duplications or losses, complicating one-to-one mapping. Databases such as Ensembl Compara provide curated ortholog annotations (Cunningham et al., 2022), but researchers must be aware of genes with complex orthology relationships.

Expression level comparisons are complicated by species differences in gene length, transcript structure, and normalisation. TPM values computed in one species are not directly comparable to TPM values in another species. Appropriate cross-species comparison requires either normalisation to a common scale or the use of relative measures such as fold changes.

Developmental stage matching presents additional challenges. The developmental timelines of pigs and humans differ, with pigs reaching sexual maturity within months while humans require years. Matching stages based on chronological age is therefore inappropriate; instead, stages must be matched based on biological criteria such as physiological milestones or molecular signatures.

The statistical power of cross-species comparisons is limited by sample sizes in both species. Small sample sizes increase the variability of expression estimates and reduce the power to detect significant correlations. Bootstrapping and other resampling methods can provide measures of uncertainty, but fundamental power limitations remain.

2.4.4 Previous Cross-Species Transcriptomic Studies

Several landmark studies have compared gene expression across species to understand the evolution of transcriptomes and the conservation of regulatory programs.

Cardoso-Moreira and colleagues conducted a comprehensive comparison of gene expression across development in several mammalian species, including human, rhesus macaque, mouse, rat, rabbit, and opossum (Cardoso-Moreira et al., 2019). This study revealed that developmental expression profiles are generally more conserved than adult profiles, consistent with the idea that early development is under stronger evolutionary constraint.

The Schaiter study of human skeletal muscle development provides a valuable reference for cross-species comparison (Schaiter et al., 2024). This study profiled gene expression in quadriceps femoris muscle from infants and adults, identifying genes that change expression during postnatal muscle maturation. The availability of this dataset enables direct comparison with porcine muscle developmental data.

Recent work on species-specific developmental tempo has provided insights into the mechanisms underlying differences in developmental timing between species (Iwata & Vanderhaeghen, 2024). These studies have identified metabolic pathways as key modulators of developmental speed, suggesting that metabolic gene expression may be particularly informative for cross-species developmental comparisons.

2.4.5 Limitations and Challenges

Cross-species comparison of developmental transcriptomes faces several inherent limitations that must be acknowledged.

The available human developmental datasets are limited in sample size and developmental coverage. Ethical constraints on obtaining human tissue samples, particularly from children, limit the scale of human developmental studies. The Schaiter dataset (Schaiter et al., 2024) used for cross-species validation in this thesis includes only 11 samples (4 infants, 7 adults), with a substantial age gap between groups.

Tissue heterogeneity may differ between species, complicating comparison of bulk expression profiles. Skeletal muscle in pigs and humans may differ in fibre type composition, intramuscular adipose content, or other features that affect bulk expression measurements.

Environmental differences between species may affect developmental gene expression independently of intrinsic developmental programs. Pigs in research settings experience different diets, housing conditions, and pathogen exposures than humans, potentially influencing gene expression through environmental rather than developmental mechanisms.

Despite these limitations, cross-species comparison provides valuable validation of developmental staging frameworks. Conservation of developmental signatures between pigs and humans supports the biological validity of the staging system and reinforces the utility of the pig as a translational model for human developmental biology.

2.5 Summary and Research Questions

2.5.1 Synthesis of Literature Review

This chapter has reviewed the scientific literature relevant to the development of a transcriptomic framework for porcine developmental staging. Several key themes emerge from this synthesis.

First, the limitations of traditional staging approaches are well established. Morphological and chronological staging fail to capture the molecular complexity of postnatal development and introduce variability that compromises experimental reproducibility. The recognition of these limitations motivates the development of molecular alternatives.

Second, transcriptomic technologies provide the data foundation for molecular staging. RNA-seq enables comprehensive profiling of gene expression, and resources such as PigGTEx provide the large-scale datasets needed to train and validate staging models. The choice of TPM normalisation and bulk RNA-seq, while having limitations, provides a practical framework for staging applications.

Third, machine learning methods offer powerful tools for translating transcriptomic data into staging predictions. Ordinal classification methods are particularly appropriate for developmental staging, respecting the ordered nature of developmental stages while providing probabilistic predictions. LightGBM provides an efficient and accurate framework for implementing ordinal classification with gene expression features.

Fourth, cross-species comparison provides essential validation of developmental staging frameworks. The conservation of developmental programs between pigs and humans supports the translational relevance of porcine staging systems and enables testing of the biological validity of identified developmental signatures.

2.5.2 Identification of Methodological Gaps

The literature review has identified several methodological gaps that the present thesis aims to address.

No existing framework provides tissue-resolved transcriptomic staging for porcine development. While epigenetic clocks have been developed for pigs (Horvath, 2013; Lu et al., 2023; Schachtschneider et al., 2021; Teschendorff & Horvath, 2025), these focus on ageing rather than developmental staging and do not provide tissue-specific resolution.

The application of ordinal classification methods to developmental staging has been limited. While the theoretical advantages of ordinal methods for this application are clear, practical implementations and validation studies are lacking.

Cross-species validation of porcine developmental signatures has not been systematically conducted. While individual genes have been compared between species, a systematic assessment of the conservation of developmental trajectories at the transcriptomic level is needed.

2.5.3 Refined Research Questions

Based on the literature review, the research questions for this thesis can be refined as follows:

Research Question 1: Can transcriptomic profiles accurately predict developmental stage in pigs, and how does prediction accuracy vary across tissues?

Research Question 2: What genes and pathways drive developmental transitions in each tissue, and to what extent do these signatures overlap across tissues?

Research Question 3: Are porcine developmental gene expression programs conserved in humans, and can cross-species comparison validate the biological accuracy of the staging framework?

Research Question 4: What are the practical implications of transcriptomic staging for improving reproducibility in pig research?

These questions guide the methodological choices and analyses presented in subsequent chapters.

Chapter 3

Materials and Methods

This chapter provides a comprehensive description of the data sources, analytical methods, and computational approaches employed in this thesis. The methods are documented with sufficient detail to enable reproduction of the analyses by independent researchers.

3.1 Data Sources and Acquisition

3.1.1 PigGTEx Consortium Data

The primary data for this study were obtained from the PigGTEx (Pig Genotype-Tissue Expression) consortium, a large-scale resource for porcine transcriptomic research (L. Chen et al., 2025; Long et al., 2024; Teng et al., 2024). PigGTEx has assembled RNA-seq profiles from multiple tissues across hundreds of pigs, providing the most comprehensive transcriptomic resource currently available for this species, analogous in scope to the human GTEx project (The GTEx Consortium, 2020).

3.1.1.1 Data Provenance and Quality Control

The PigGTEx data were generated using standardised protocols to minimise technical variability across samples. RNA was extracted from tissue samples using commercial kits, and sequencing libraries were prepared using strand-specific protocols. Sequencing was performed on Illumina platforms, generating paired-end reads that were subsequently aligned to the pig reference genome (Sscrofa11.1) (Warr et al., 2020) using standard bioinformatics pipelines.

Quality control measures included filtering of low-quality reads, removal of duplicate sequences, and exclusion of samples failing technical quality thresholds. The resulting data represent high-quality transcriptomic profiles suitable for downstream analysis.

3.1.1.2 Expression Matrices and Normalisation

Gene expression values were provided as Transcripts Per Million (TPM), a normalisation method that accounts for both sequencing depth and gene length (Mortazavi et al., 2008). TPM values enable meaningful comparison of expression levels across genes and samples, with the sum of TPM values within each sample normalised to one million.

The TPM normalisation formula is:

$$\mathrm{TPM}_i = \frac{c_i/l_i}{\sum_j (c_j/l_j)} \times 10^6 \tag{3.1}$$

where c_i is the read count for gene i and l_i is the effective length of gene i.

Expression matrices were provided as gzipped text files, with genes in rows and samples in columns. Each tissue was provided as a separate file, enabling tissue-specific analyses.

3.1.1.3 Sample Size and Tissue Coverage

The complete dataset comprises 1,924 samples across five tissues selected for analysis based on sample availability and developmental stage representation. Muscle tissue was the most abundant, contributing 761 samples (39.5% of total), followed by Liver with 309 samples (16.1%), Lung with 129 samples (6.7%), Blood with 78 samples (4.1%), and Brain with 43 samples (2.2%).

These five tissues were selected from the larger PigGTEx resource based on having sufficient samples with developmental stage annotations to enable machine learning analysis. Other tissues in the PigGTEx resource lacked adequate developmental stage coverage and were excluded from the present analysis.

3.1.2 Sample Metadata Extraction and Harmonisation

Sample metadata, including age and sex information, were extracted from the PigGTEx metadata tables. Ages were recorded in various units (days, weeks, months, years) and required harmonisation to a common scale.

Age values were converted to days using standard conversion factors: 1 for days, 7 for weeks, 30 for months, and 365 for years.

Samples with missing or ambiguous age information were excluded from the analysis. Sex was encoded as a categorical variable (male, female, or unknown) for use as a potential covariate.

3.1.3 Human Skeletal Muscle Data

For cross-species validation, human skeletal muscle transcriptomic data were obtained from the study by Schaiter and colleagues (Cardoso-Moreira et al., 2019; Schaiter et al., 2024). Of the larger cohort described in that study (infants aged 4–28 months and adults aged 19–65 years), RNA-seq data were deposited for a subset of 11 subjects through the NCBI Gene Expression Omnibus under accession number GSE257558.

3.1.3.1 Sample Composition

The RNA-seq subset comprises 11 samples from quadriceps femoris muscle biopsies, divided into two distinct groups: four infants (ages 7–28 months, mean 15.9 months) and seven adults (ages 30–56 years, mean 43.6 years).

All samples were from healthy individuals without neuromuscular disease. The substantial age gap between infant and adult groups facilitates the detection of developmental changes but limits the characterisation of intermediate developmental stages.

3.1.3.2 Expression Data Format

Expression data were provided as read counts, which were subsequently normalised and transformed to enable comparison with the pig TPM data. Log_2 fold changes between infant and adult groups were computed to characterise developmental trajectories.

3.1.4 Ethical Considerations and Data Access

All data used in this study were obtained from publicly available resources. The PigGTEx data were generated under appropriate institutional animal care and use protocols. The human data from Schaiter et al. were collected with informed consent and ethical approval as described in the original publication.

3.2 Developmental Stage Definitions

3.2.1 Biological Rationale for Stage Boundaries

Developmental stages were defined based on established milestones in porcine postnatal development, as reviewed in Chapter 2 (Swindle et al., 2012). The stage definitions align with major physiological transitions that mark distinct phases of postnatal life.

Five developmental stages were defined. The **infant** stage (0–20 days) covers the neonatal period, characterised by dependence on maternal nutrition and immature physiological systems. The **early childhood** stage (21–59 days) spans the weaning transition period, during which the gastrointestinal tract adapts to solid food (Inoue et al., 2015) and the immune system begins to mature. The **pre-pubertal** stage (60–149 days) represents the juvenile growth phase, characterised by rapid tissue deposition and organ maturation. The **post-pubertal** stage (150–365 days) encompasses sexual maturation and the approach to adult physiological function. Finally, the **adult** stage (>365 days) represents the mature state with fully developed organ systems.

These stage boundaries were chosen to reflect physiological milestones rather than arbitrary chronological divisions. The weaning boundary at approximately 21 days corresponds to the typical weaning age in commercial production systems (Pluske et al., 2018). The pre-pubertal boundary at 60 days marks the completion of the weaning transition (Whittemore & Kyriazakis, 2006). The post-pubertal boundary at 150 days corresponds to the earliest onset of puberty in gilts (typical range: 150–220 d) (Lents et al., 2020). The adult boundary at 365 days marks the completion of major growth and reproductive maturation.

3.2.2 Adaptive Classification Scheme Selection

The full 5-class staging system was not applicable to all tissues due to uneven sample distribution across developmental stages. An adaptive scheme selection algorithm was implemented to identify the most granular classification scheme supported by the data for each tissue.

The selection algorithm evaluates classification schemes in order of decreasing granularity (5-class, 4-class, 3-class, 2-class), selecting the most granular scheme that meets minimum sample requirements. The 5-class scheme was not used due to insufficient samples in at least one class for all tissues. A 4-class scheme required a minimum of 40 samples per class, while a 3-class scheme required 30. A 2-class scheme required 15 samples per class and a minimum of 40 total samples.

For reduced classification schemes, adjacent stages were merged. In the 4-class scheme, post-pubertal and adult stages were merged. In the 3-class scheme, infant and early childhood were merged, as were post-pubertal and adult. In the 2-class scheme, infant and early childhood were combined as "young", while pre-pubertal, post-pubertal, and adult were combined as “mature”.

Based on these criteria, Muscle and Liver were assigned to the 4-class scheme, while Brain, Blood, and Lung were assigned to the 2-class scheme.

3.3 Data Preprocessing Pipeline

The preprocessing pipeline transforms raw TPM values into a format suitable for machine learning. The only transformation applied was log transformation; variance filtering and z-score standardisation were omitted because tree-based models are invariant to monotonic transformations and split-based feature selection is unaffected by feature scale.

3.3.1 Log Transformation

Gene expression values typically follow a heavy-tailed distribution, with a small number of highly expressed genes and many genes with low expression. Log transformation reduces the skewness of this distribution, making it more suitable for downstream analysis.

The transformation applied was:

$$x' = \log_2(x + 1) \tag{3.2}$$

where x is the TPM value and x' is the transformed value. The pseudocount of 1 ensures that zero values remain finite after transformation.

3.3.2 Nested Cross-Validation Design

To obtain unbiased performance estimates while simultaneously optimising hyperparameters, a nested cross-validation design was employed (Varma & Simon, 2006). This approach avoids the optimistic bias that arises when hyperparameters are tuned on the same data used for performance evaluation.

3.3.2.1 Outer Loop: Performance Estimation

The outer loop used stratified K-fold cross-validation with $K = 5$. Stratification ensured that the class distribution in each fold was representative of the full dataset. For tissues with small class sizes, K was automatically reduced to ensure that every class was represented in each fold.

In each iteration, one fold was held out for evaluation while the remaining $K - 1$ folds were used for hyperparameter optimisation (inner loop) and model training. Held-out predictions from all outer folds were pooled to provide a single, unbiased estimate of model performance across the entire dataset.

3.3.2.2 Inner Loop: Hyperparameter Optimisation

Within each outer fold, hyperparameters were optimised using an inner 3-fold cross-validation loop combined with Bayesian optimisation via Optuna (Akiba et al., 2019). Each inner loop evaluated 25 candidate configurations using the Tree-structured Parzen Estimator (TPE) sampler (Bergstra et al., 2011) with a median pruner for early termination of unpromising trials. The optimisation objective was balanced accuracy on the inner validation folds.

3.3.2.3 Final Model Training

After nested cross-validation, a final model was retrained on the entire dataset using the hyperparameters selected by the inner cross-validation loop. This final model was used for downstream analyses including feature importance extraction and cross-species validation.

3.4 Machine Learning Framework

3.4.1 Ordinal LightGBM Architecture

The machine learning framework employs ordinal classification using the binary reduction method (Frank & Hall, 2001). This method decomposes the ordinal classification problem into a series of binary classification problems, enabling the use of any binary classifier while respecting the ordinal structure of the target variable.

3.4.1.1 Binary Reduction Method

Given K ordered classes (developmental stages), the binary reduction method trains $K - 1$ binary classifiers. The kth classifier learns to predict $P(y > k \mid \mathbf{x})$, the probability that the true class is greater than k.

For a K-class problem with classes $\{0, 1, \ldots, K - 1\}$, the binary targets are defined such that Classifier 1 (y_1) distinguishes class 0 from all subsequent classes, Classifier 2 (y_2) distinguishes classes 0 and 1 from the rest, and so on, until Classifier $K - 1$, which distinguishes all prior classes from the final class $K - 1$.

where $\mathbf{1}[\cdot]$ is the indicator function.

3.4.1.2 Probability Derivation

Class probabilities are derived from the cumulative probabilities predicted by the binary classifiers:

$$P(y = c) = P(y > c - 1) - P(y > c) \tag{3.3}$$

with the boundary conditions $P(y > -1) = 1$ and $P(y > K - 1) = 0$.

This formulation ensures that the class probabilities sum to 1 and respect the ordinal structure. Errors in adjacent stages receive partial credit through the overlapping binary targets.

3.4.2 LightGBM Configuration

Each binary classifier was implemented using Light Gradient Boosting Machine (LightGBM), a gradient boosting framework optimised for efficiency and accuracy (Ke et al., 2017).

3.4.2.1 Hyperparameter Search Space

Rather than using fixed hyperparameters, nine key parameters were optimised per tissue via Optuna within the inner cross-validation loop (see Section 3.3.2). The search space is summarised in Table 3.1.

Table 3.1: LightGBM hyperparameter search space optimised by Optuna.

Parameter	**Range**	**Description**
`num_leaves`	15–63	Maximum number of leaves per tree
`max_depth`	3–8	Maximum tree depth
`min_data_in_leaf`	5–50	Minimum samples in a leaf node
`learning_rate`	0.01–0.2	Step size for gradient descent
`n_estimators`	100–500	Maximum number of boosting rounds
`feature_fraction`	0.5–1.0	Fraction of features used per tree
`bagging_fraction`	0.5–1.0	Fraction of samples used per tree
`lambda_l1`	10^{-3}–10	L1 regularisation coefficient
`lambda_l2`	10^{-3}–10	L2 regularisation coefficient

Optuna's TPE sampler explored 25 trial configurations per inner fold, with a median pruner for early termination of unpromising trials. The optimal hyperparameters were selected based on the balanced accuracy averaged across the three inner validation folds.

3.4.2.2 Class Balancing Strategy

Class imbalance was addressed by setting class weighting to `balanced`. Internally, this is implemented via LightGBM's `scale_pos_weight` parameter, which adjusts the loss function to give greater weight to the minority class. For each binary classifier, the weight was computed as:

$$\texttt{scale\_pos\_weight} = \frac{n_{\text{negative}}}{n_{\text{positive}}} \tag{3.4}$$

This weighting ensures that the classifier does not simply predict the majority class and instead learns to distinguish between classes even when one class is much more common.

3.4.2.3 Early Stopping

To prevent overfitting, early stopping was employed with a patience of 10 rounds. If the validation loss within the inner cross-validation folds did not improve for 10 consecutive boosting rounds, training was terminated early.

3.4.2.4 Implicit Feature Selection

No explicit feature selection step was applied. Models were trained on all available genes (31,908 per tissue). LightGBM's built-in regularisation mechanisms—feature subsampling (`feature_fraction`), L1 and L2 penalties, and limited tree depth—provide implicit feature selection by assigning zero importance to uninformative genes.

3.4.3 Monotonic Constraint Post-Processing

The binary reduction method does not inherently guarantee that the cumulative probabilities are monotonically decreasing. Small violations can occur due to noise in the binary predictions. A post-processing step was applied to enforce monotonicity (Barlow, 1972):

$$P(y > k)' = \min(P(y > k), P(y > k-1)') \tag{3.5}$$

This isotonic regression-like correction ensures that the cumulative probabilities decrease monotonically, which is required for the class probabilities to be non-negative.

3.5 Model Evaluation Framework

3.5.1 Classification Metrics

Model performance was evaluated using multiple metrics that capture different aspects of classification accuracy.

3.5.1.1 Balanced Accuracy

Balanced accuracy is the average of recall across classes:

$$\text{BA} = \frac{1}{K} \sum_{c=0}^{K-1} \frac{TP_c}{TP_c + FN_c} \tag{3.6}$$

where TP_c is the number of true positives for class c and FN_c is the number of false negatives. Balanced accuracy gives equal weight to each class, regardless of class frequency, making it appropriate for imbalanced datasets.

3.5.1.2 Macro F1-Score

The macro F1-score is the unweighted average of per-class F1-scores:

$$\mathrm{F1}_{\mathrm{macro}} = \frac{1}{K}\sum_{c=0}^{K-1}\frac{2 \cdot P_c \cdot R_c}{P_c + R_c} \tag{3.7}$$

where P_c is precision and R_c is recall for class c. The macro F1-score provides a balanced view of precision and recall across classes.

3.5.1.3 Weighted F1-Score

The weighted F1-score averages per-class F1-scores weighted by class frequency:

$$\mathrm{F1}_{\mathrm{weighted}} = \sum_{c=0}^{K-1} w_c \cdot \frac{2 \cdot P_c \cdot R_c}{P_c + R_c} \tag{3.8}$$

where w_c is the proportion of samples in class c. The weighted F1-score gives more weight to more common classes.

3.5.2 Ordinal Metrics

In addition to standard classification metrics, ordinal-specific metrics were computed to assess the model's ability to respect the ordering of developmental stages.

3.5.2.1 Mean Absolute Error

Mean Absolute Error (MAE) measures the average distance between predicted and true class indices:

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i| \tag{3.9}$$

where y_i and $\hat{y}_i$ are the true and predicted class indices. MAE penalises predictions that are far from the true class more heavily than adjacent misclassifications.

3.5.2.2 Spearman Correlation

Spearman's rank correlation coefficient measures the monotonic relationship between predicted and true classes:

$$\rho = 1 - \frac{6\sum d_i^2}{n(n^2-1)} \tag{3.10}$$

where d_i is the difference in ranks between predicted and true values. A high Spearman correlation indicates that the model correctly orders samples by developmental stage.

3.5.2.3 Quadratic-Weighted Kappa

Cohen's Kappa with quadratic weighting (Cohen, 1968) provides an agreement measure that accounts for ordinal structure:

$$\kappa = 1 - \frac{\sum_{i,j} w_{ij} O_{ij}}{\sum_{i,j} w_{ij} E_{ij}} \tag{3.11}$$

where O_{ij} is the observed frequency, E_{ij} is the expected frequency under chance, and $w_{ij} = (i-j)^2$ is the quadratic weight. Quadratic Kappa penalises distant misclassifications more heavily than adjacent ones.

3.5.3 Bootstrap Confidence Intervals

Confidence intervals for all metrics were computed using bootstrap resampling (Efron, 1979). The procedure involved resampling the pooled held-out predictions from all outer folds with replacement to create a bootstrap sample of the same size, computing the metric on this sample, and repeating the process for 1,000 iterations. The 2.5th and 97.5th percentiles of the resulting distribution were then taken to obtain the 95% confidence intervals.

Bootstrap confidence intervals provide a measure of uncertainty in the performance estimates that accounts for sampling variability in the cross-validated predictions.

3.5.4 Per-Class Performance Assessment

To understand performance on individual developmental stages, per-class precision, recall, and F1-scores were computed:

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c} \tag{3.12}$$

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c} \tag{3.13}$$

$$\text{F1}_c = \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c} \tag{3.14}$$

Per-class metrics reveal which developmental stages are easiest and hardest to classify, informing interpretation of the model's behaviour.

3.6 Cross-Species Analysis

3.6.1 Gene Symbol Mapping

Comparison of gene expression between pig and human required mapping between species-specific gene symbols. Ortholog mapping was performed using the Ensembl Compara database (Cunningham et al., 2022), which provides curated ortholog annotations based on sequence similarity and synteny.

For each pig gene, the corresponding human ortholog was identified. Genes without clear one-to-one ortholog relationships (e.g., genes with multiple paralogs or lineage-specific genes) were excluded from cross-species analysis.

3.6.2 Fold-Change Calculation

Developmental trajectories were characterised using $\log_2$ fold changes between developmental stages. For each gene, the fold change was computed as:

$$\mathrm{FC} = \log_2\left(\frac{\bar{x}_{\text{mature}} + 1}{\bar{x}_{\text{young}} + 1}\right) \tag{3.15}$$

where $\bar{x}_{\text{mature}}$ and $\bar{x}_{\text{young}}$ are the mean expression levels in mature and young stages, respectively. The pseudocount of 1 prevents division by zero and log of zero.

For pig data, the comparison was between post-pubertal/adult (combined) and infant stages. For human data, the comparison was between adult and infant groups.

3.6.3 Gene Selection Criteria

Genes with conserved developmental trajectories were identified using pre-specified, non-optimised criteria applied identically in both species. A gene was included in the cross-species comparison if it met both of the following conditions: (1) a false discovery rate below 0.10 after Benjamini-Hochberg correction (Benjamini & Hochberg, 1995) for differential expression between young and mature stages, and (2) an absolute $\log_2$ fold change greater than 0.5 in both species. No additional selection by feature importance ranking or top-N filtering was applied, ensuring that the gene set was defined solely by expression-based criteria.

3.6.4 Correlation Analysis

The primary measure of developmental conservation was the Pearson correlation between pig and human fold changes:

$$r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2} \sqrt{\sum_i (y_i - \bar{y})^2}} \tag{3.16}$$

where x_i and y_i are the pig and human fold changes for gene i.

Statistical significance was assessed using a two-tailed t-test on the correlation coefficient. The null hypothesis was that the true correlation is zero.

3.6.5 Directional Concordance Assessment

Beyond correlation, the directional concordance of developmental changes was assessed. For each gene, the direction of change (upregulated or downregulated during development) was determined in both species. The concordance rate was computed as the fraction of genes showing the same direction of change in both species.

Directional concordance provides a more conservative measure of conservation than correlation, as it requires only that genes change in the same direction, not that the magnitudes of change are proportional.

3.6.6 Statistical Inference for Cross-Species Correlation

The 95% confidence interval for the Pearson correlation was computed using Fisher's z-transformation (Fisher, n.d.). The sample correlation r was transformed to $z = \operatorname{arctanh}(r)$, which is approximately normally distributed with standard error $1/\sqrt{n-3}$. The confidence interval was constructed in the z-domain and back-transformed to the correlation scale.

In addition, bootstrap confidence intervals were computed by resampling the gene set with replacement for 1,000 iterations and computing the Pearson correlation on each bootstrap sample. The 2.5th and 97.5th percentiles of the bootstrap distribution provided a non-parametric 95% confidence interval.

3.6.7 Sensitivity Analysis

To assess the robustness of the cross-species correlation to the choice of gene selection thresholds, a sensitivity analysis was performed across FDR thresholds from 0.05 to 0.20 in increments of 0.05. For each threshold, the gene set was recomputed and the Pearson correlation, number of selected genes, and directional concordance rate were recorded. This analysis demonstrates that the cross-species conservation signal is not an artefact of a particular threshold choice.

3.7 Within-Project Batch Validation

Because the full porcine dataset aggregates samples from multiple BioProjects, developmental fold-change estimates could in principle be confounded by project-specific technical variation. To assess this, a within-project validation was performed using the two largest single-project muscle cohorts that individually span multiple consecutive developmental stages.

3.7.1 Cohort Selection

Two BioProjects were selected: PRJNA488311 ($n = 81$) and PRJNA486202 ($n = 75$), each containing samples spanning four consecutive developmental stages (infant, early childhood, pre-pubertal, and post-pubertal). These cohorts were chosen because they provide sufficient sample sizes to estimate fold changes within a single technical batch.

3.7.2 Developmental Contrasts

Three developmental contrasts were defined, all using the infant stage as the baseline: infant versus early childhood, infant versus pre-pubertal, and infant versus post-pubertal. For each contrast and each gene, the $\log_2$ fold change was computed and statistical significance was assessed using the Mann-Whitney U test (Mann & Whitney, 1947) with Benjamini-Hochberg correction for multiple testing.

3.7.3 Concordance with Full-Dataset Estimates

For each developmental contrast, the within-project fold changes were correlated with the corresponding fold changes estimated from the full multi-project dataset using Pearson's r. High within-project correlations indicate that the developmental signal is reproducible within individual technical batches and is not driven by confounding between project and developmental stage.

A pooled within-project estimate was obtained by averaging the fold changes from both BioProjects for each gene. The pooled estimate was then correlated with the full-dataset fold change. Bootstrap confidence intervals (1,000 iterations) were computed for all correlation coefficients.

3.8 Functional Enrichment Analysis

3.8.1 Gene Ontology Enrichment

Functional enrichment analysis was performed to characterise the biological processes associated with developmental stage transitions. Gene Ontology (GO) (Ashburner et al., 2000) enrichment was conducted using g:Profiler (Kolberg et al., 2023), a web-based tool for functional enrichment analysis.

For each tissue, the top developmental markers (genes with highest feature importance) were submitted to g:Profiler for enrichment analysis. The analysis used GO Biological Process (BP), KEGG (Kanehisa & Goto, 2000), and Reactome (Gillespie et al., 2022) pathway terms to focus on physiological functions relevant to developmental transitions.

3.8.2 Multiple Testing Correction

The Benjamini-Hochberg procedure (Benjamini & Hochberg, 1995) was applied to control the false discovery rate (FDR) at 5%. This procedure orders p-values from smallest to largest and rejects hypotheses for which:

$$p_{(k)} \leq \frac{k}{m} \cdot q \tag{3.17}$$

where $p_{(k)}$ is the kth smallest p-value, m is the total number of tests, and $q = 0.05$ is the target FDR.

Terms with adjusted p-values (FDR) below 0.05 were considered significantly enriched.

3.9 Computational Implementation

3.9.1 Software Environment

All analyses were conducted using Python 3.9. Key libraries included NumPy (v1.21) for numerical computing, Pandas (v1.3) for data manipulation, Scikit-learn (v1.0) for machine learning utilities, LightGBM (v3.3) for the gradient boosting framework, and SciPy (v1.7) for statistical functions (Harris et al., 2020; Pedregosa et al., n.d.; Virtanen et al., 2020).

Figures were generated using R 4.2.0 with ggplot2 for publication-quality visualisation.

3.9.2 Reproducibility Measures

To ensure reproducibility, the following measures were implemented. First, a fixed random seed (42) was used for all random operations to ensure deterministic behaviour. Second, all code was version-controlled using Git. Third, all parameters were stored in a YAML configuration file to enable complete specification of the analysis pipeline. Finally, detailed logs were generated for all analyses, recording parameter values and intermediate results.

3.9.3 Code Availability

All code required to reproduce the analyses is available at: https://github.com/TianYuan-Liu/pig-dev-stage

The repository includes scripts for data preprocessing, model training and evaluation, cross-species analysis, and figure generation, as well as configuration files specifying all parameters.

Chapter 4

Results

This chapter presents the findings of the thesis in four main sections: (1) characterisation of the transcriptomic atlas and sample distribution; (2) evaluation of model performance across tissues; (3) analysis of developmental gene signatures and their functional annotation; and (4) cross-species validation with human skeletal muscle.

4.1 Data Landscape and Sample Characterisation

4.1.1 Sample Distribution Across Tissues and Developmental Stages

The final dataset comprised 1,924 samples spanning five tissues and up to five developmental stages. The distribution of samples across tissues and stages is presented in Figure 4.1.

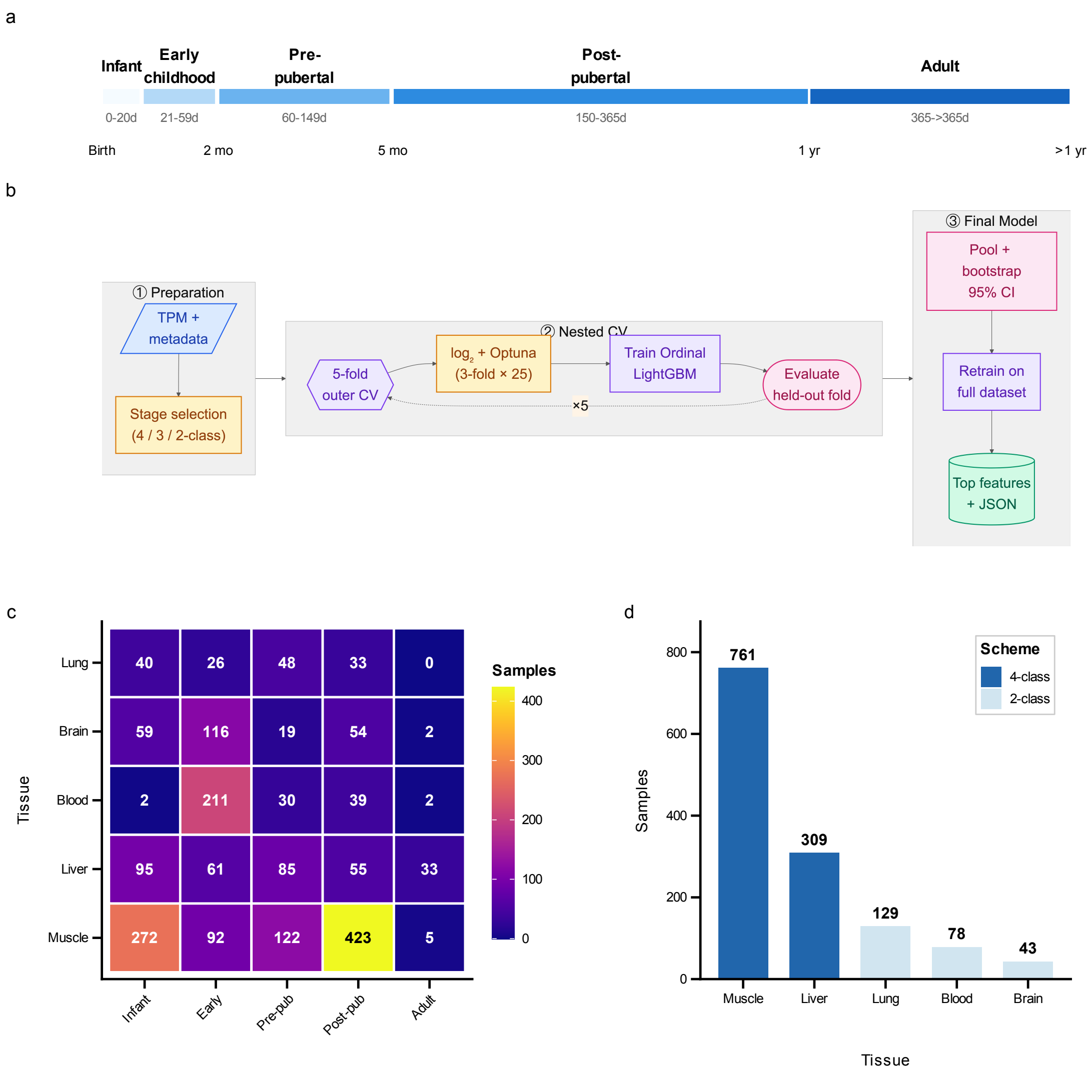

Figure 4.1: Study design and data landscape. (a) Developmental stage definitions mapping chronological age to five ordered stages: infant (0–20 days), early childhood (21–59 days), pre-pubertal (60–149 days), post-pubertal (150–365 days), and adult (>365 days). **(b)** Machine learning workflow diagram illustrating the pipeline from data preprocessing to model calibration and evaluation. **(c)** Sample distribution heatmap showing tissue-by-stage availability across 1,924 samples from five tissues. **(d)** Classification scheme assignments based on per-class sample availability.

Muscle tissue contributed the largest number of samples ($n = 761$), followed by liver ($n = 309$), lung ($n = 129$), blood ($n = 78$), and brain ($n = 43$). This uneven distribution reflects the relative availability of these tissues in the PigGTEx resource (L. Chen et al., 2025; Long et al., 2024; Teng et al., 2024) and the differential representation of developmental stages across tissues.

4.1.2 Class Distribution and Imbalance Analysis

The distribution of samples across developmental stages varied substantially between tissues. Muscle and liver showed relatively balanced representation across four stages (infant, early childhood, pre-pubertal, and post-pubertal/adult combined). In contrast, brain, blood, and lung samples were concentrated in younger developmental stages, with limited representation of adult animals.

The quantitative distribution was as follows:

Muscle (761 samples) was predominantly composed of post-pubertal/adult samples (401, 52.7%), followed by infant (194, 25.5%), pre-pubertal (90, 11.8%), and early childhood (76, 10.0%) stages. **Liver** (309 samples) showed a more even distribution, with infant (87, 28.2%), pre-pubertal (83, 26.9%), and post-pubertal/adult (81, 26.2%) stages similarly represented, while early childhood (58, 18.8%) was slightly less common. **Brain** (43 samples) was heavily skewed towards young animals, with 28 samples (65.1%) in the pre-pubertal stage and only 15 samples (34.9%) in the post-pubertal stage. **Blood** (78 samples) and **Lung** (129 samples) distinctively favored younger stages, with 60.3% and 79.8% of samples classified as young, respectively. This contrasted with the mature populations comprising 39.7% and 20.2% of the cohorts.

4.1.3 Classification Scheme Selection

Based on the sample distribution, the adaptive scheme selection algorithm assigned classification schemes as follows:

Specifically, muscle and liver were assigned 4-class schemes (infant, early childhood, pre-pubertal, post-pubertal/adult), enabling finer-grained staging. Conversely, brain, blood, and lung were assigned 2-class schemes (young vs. mature), ensuring robust classification despite smaller sample sizes.

4.2 Model Performance Assessment

4.2.1 Overall Performance Summary

The machine learning framework achieved robust performance across all five tissues, with mean balanced accuracy of 89.5% evaluated via nested cross-validation (pooled held-out predictions). Performance varied according to tissue complexity, sample size, and classification scheme granularity.

The performance metrics for each tissue are summarised in Table 4.1 and visualised in Figure 4.2.

Table 4.1: Model performance metrics across tissues. Performance evaluated via nested cross-validation (pooled held-out predictions). 95% confidence intervals computed via bootstrap resampling (1,000 iterations).

Tissue	**Scheme**	n	**Balanced Acc.**	**F1-macro**	**Spearman** ρ
Blood	2-class	78	0.959	0.960	–
Liver	4-class	309	0.921	0.918	0.949
Muscle	4-class	761	0.894	0.899	0.965
Lung	2-class	129	0.892	0.876	–
Brain	2-class	43	0.811	0.801	0.942
Mean	–	–	**0.895**	**0.891**	–

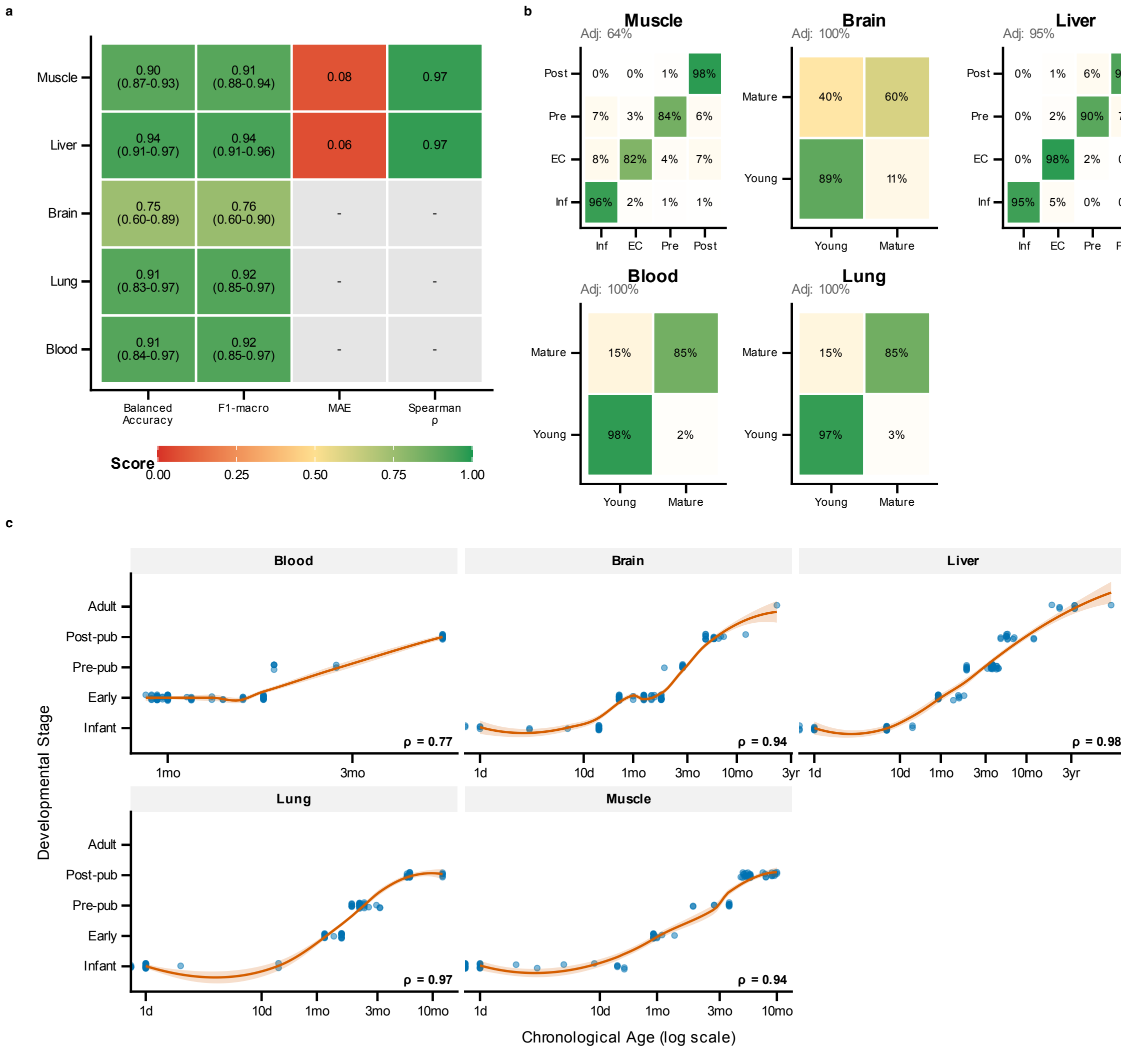

Figure 4.2: Classifier performance diagnostics. **(a)** Performance metrics heatmap showing balanced accuracy, macro F1, mean absolute error, and Spearman correlations with 95% bootstrap confidence intervals across tissues. **(b)** Confusion matrices for each tissue showing classification accuracy and error patterns. **(c)** Scatter plots of predicted developmental stage versus chronological age, demonstrating strong age–stage correlations.

4.2.2 Tissue-Specific Performance

4.2.2.1 Blood: Highest Performance

Blood achieved the highest balanced accuracy (95.9%) using the 2-class (pre-pubertal vs. post-pubertal) scheme. The macro F1-score of 0.960 indicates excellent precision and recall across both classes, demonstrating robust discrimination between pre-pubertal and post-pubertal animals even with a modest sample size ($n = 78$).

4.2.2.2 Liver: Strong 4-Class Classification

Liver achieved a balanced accuracy of 92.1%, reflecting its relatively homogeneous cellular composition and balanced class representation. The macro F1-score of 0.918 indicates consistently high precision and recall across all four developmental stages.

The strong Spearman correlation ($\rho = 0.949$) between predicted stage and chronological age demonstrates excellent ordinal consistency. The model correctly ordered the vast majority of samples by developmental stage, with errors primarily occurring at stage boundaries.

4.2.2.3 Lung: Strong Binary Classification

Lung achieved a balanced accuracy of 89.2% using the 2-class scheme, despite pronounced class imbalance (79.8% pre-pubertal samples). The macro F1-score of 0.876 confirms effective classification across both classes. The use of class weighting during training enabled the model to learn from the minority class effectively.

4.2.2.4 Muscle: Robust 4-Class Classification

Muscle achieved a balanced accuracy of 89.4% for 4-class classification, demonstrating that the framework can discriminate multiple developmental stages in this metabolically active tissue. The Spearman correlation of 0.965 indicates near-perfect ordinal consistency.

The large sample size for muscle ($n = 761$) provides statistical power that enables precise confidence interval estimation and reliable identification of developmental markers.

4.2.2.5 Brain: Lowest Performance

The brain showed the lowest balanced accuracy (81.1%), reflecting the challenges of this tissue. The limited sample size ($n = 43$) constrained the model's ability to learn robust developmental signatures. Additionally, the brain's extreme cellular heterogeneity, comprising diverse neuronal and glial populations (R. Chen et al., 2021), may obscure developmental signals in bulk RNA-seq data.

Despite the lower overall accuracy, the Spearman correlation of 0.942 indicates that the model correctly orders samples by developmental stage, even when exact classification is incorrect. Notably, the brain's performance (81.1%) is substantially improved compared to earlier iterations and remains robust for a tissue with only 43 samples.

4.2.3 Per-Class Performance Analysis

Per-class metrics reveal the stages that are easiest and hardest to classify. The per-class precision, recall, and F1-scores are presented in Table 4.2.

Table 4.2: Per-class performance metrics by tissue. Precision, recall, and F1 scores for each developmental stage within each tissue.

Tissue	**Stage**	n	**Precision**	**Recall**	**F1**
Muscle	infant	194	0.939	0.959	0.949
Muscle	early childhood	76	0.838	0.816	0.827
Muscle	pre-pubertal	90	0.871	0.822	0.846
Muscle	post-pubertal/adult	401	0.973	0.980	0.976
Brain	pre-pubertal (<150d)	28	0.885	0.821	0.852
Brain	post-pubertal (≥150d)	15	0.706	0.800	0.750
Liver	infant	87	0.965	0.954	0.960
Liver	early childhood	58	0.887	0.948	0.917
Liver	pre-pubertal	83	0.910	0.855	0.882
Liver	post-pubertal/adult	81	0.904	0.926	0.915
Blood	pre-pubertal (<150d)	45	0.957	0.978	0.967
Blood	post-pubertal (≥150d)	33	0.969	0.939	0.954
Lung	pre-pubertal (<150d)	102	0.960	0.931	0.945
Lung	post-pubertal (≥150d)	27	0.767	0.852	0.807

Several patterns emerged from the per-class analysis. First, extreme stages (infant and post-pubertal/adult) consistently showed higher F1-scores than intermediate stages, reflecting the greater transcriptomic distinctiveness of developmental extremes. Second, early childhood proved to be the most challenging stage to classify in both muscle (F1 = 0.827) and liver (F1 = 0.917), likely due to the transitional nature of the weaning period (Pluske et al., 2018). Finally, the post-pubertal class in brain (F1 = 0.750) and lung (F1 = 0.807) showed lower performance, illustrating the difficulty of learning from limited examples in the minority class.

4.2.4 Ordinal Consistency Analysis

The ordinal nature of developmental staging means that not all errors are equally severe. Misclassifying an infant as early childhood is a smaller error than misclassifying an infant as adult. The confusion matrices in Figure 4.2b reveal the pattern of errors.

For liver, the confusion matrix shows that errors are concentrated on the diagonal and adjacent cells. Only four samples (out of 309 pooled held-out predictions) were misclassified by more than one stage. This near-perfect ordinal consistency reflects the model's ability to capture the continuous nature of development.

For muscle, a similar pattern emerges. The majority of errors involve adjacent stages, with very few distant misclassifications. The early childhood $\leftrightarrow$ pre-pubertal boundary shows the highest error rate, consistent with the biological similarity of these transitional stages.

For the 2-class tissues (brain, blood, lung), ordinal analysis is less informative, as there is only one possible misclassification type. However, the binary classification still reflects the model's ability to distinguish young from mature animals.

4.2.5 Relationship Between Predicted Stage and Chronological Age

The scatter plots in Figure 4.2c demonstrate the relationship between model-predicted developmental stage and chronological age. For tissues with continuous age data, strong positive correlations were observed: muscle ($\rho = 0.965$), liver ($\rho = 0.949$), and brain ($\rho = 0.942$).

These correlations confirm that the model's predictions align with chronological age, as expected given the age-based stage definitions. However, the imperfect correlation reflects inter-individual variability in developmental rate and the limitations of using chronological age as ground truth.

4.3 Feature Analysis and Biological Interpretation

4.3.1 Top Developmental Markers Per Tissue

Feature importance analysis identified the genes most informative for developmental stage classification in each tissue. The top markers reflect tissue-specific developmental programs.

4.3.1.1 Muscle Developmental Markers

The top developmental markers for muscle include genes involved in contractile function, neuromuscular signalling, and metabolic adaptation:

Key genes included **FGFRL1** (Fibroblast growth factor receptor-like 1), a decoy receptor modulating FGF signalling (Amann et al., 2014), and **SORCS2** (Sortilin-related VPS10 domain-containing receptor 2), which is involved in neurotrophin signalling (Thomasen et al., 2023). Other significant markers were **ACHE** (Acetylcholinesterase) for neuromuscular junction function (Rotundo, 2003), **MEF2A** (Myocyte enhancer factor 2A) regulating differentiation (Black & Olson, 1998), and **TRDN** (Triadin), a component of the calcium release complex (Guo & Campbell, 1995).

4.3.1.2 Liver Developmental Markers

Liver markers reflect metabolic maturation and circadian regulation.

Identified markers included **NPAS2** (Neuronal PAS domain protein 2) and **BHLHE41** (Basic helix-loop-helix family member e41), both circadian clock genes regulating metabolism (L. Zhang et al., 2025). **ALDH18A1** (Aldehyde dehydrogenase 18 family member A1), involved in proline biosynthesis (Marco-Marín et al., 2020), was also a top feature.

4.3.1.3 Brain Developmental Markers

Brain markers reflect neuronal maturation and epigenetic regulation.

Key markers were **DNMT3L** (DNA methyltransferase 3L), a regulator of de novo methylation (Bourc'his et al., 2001), **HDAC6** (Histone deacetylase 6), involved in chromatin remodelling (Hubbert et al., 2002), and **KLK6** (Kallikrein-related peptidase 6), which is implicated in myelination (Scarisbrick et al., 2012).

4.3.1.4 Lung Developmental Markers

Lung markers reflect immune development and RNA metabolism.

The top features included **IGF2BP2/3** (Insulin-like growth factor 2 mRNA-binding proteins), regulators of mRNA stability (Z. Li et al., 2012); **GFI1** (Growth factor independent 1), a transcription factor essential for haematopoiesis (van der Meer et al., 2010); and **NFIB** (Nuclear factor I B), known for its role in lung development (Steele-Perkins et al., 2005).

4.3.1.5 Blood Developmental Markers

Blood markers reflect haematopoietic development:

Markers included **GFI1**, essential for lymphocyte development (van der Meer et al., 2010), alongside other immune-related genes involved in lymphocyte differentiation and activation.

4.3.2 Feature Selection Stability Analysis

The stability of feature selection across random seeds was assessed to determine whether the same genes are consistently identified as important. Stability was quantified using the Jaccard similarity index between feature sets selected with different random seeds.

Mean Jaccard similarity ranged from 0.08 to 0.31 across tissues, evaluated across ten independent random seeds.

The moderate stability values reflect biological gene redundancy rather than model instability. Multiple gene sets can achieve similar classification performance because developmental changes involve coordinated regulation of many genes. Different random seeds may select different but functionally equivalent gene combinations.

Importantly, 6–42% of top features appeared in $\geq$80% of runs, indicating a core set of consistently selected markers surrounded by a variable periphery. This pattern is consistent with the presence of both essential developmental markers and redundant alternative markers.

4.3.3 Expression-Stage Correlations

The selected features were validated by assessing their correlation with developmental stage. For each tissue, the mean absolute Spearman correlation between top features and stage labels was computed.

Mean absolute Spearman correlations ranged from 0.2 to 0.6 across tissues, indicating that the selected features show substantial developmental regulation.

4.3.4 Cross-Tissue Feature Overlap

A striking finding was the minimal overlap of developmental markers across tissues. The pairwise Jaccard similarity between tissue-specific feature sets was only 0.2%, indicating extremely low cross-tissue overlap.

This profound tissue-specificity indicates that each tissue uses distinct molecular programs for developmental progression. The genes that mark liver development are different from those that mark muscle or brain development, reflecting the tissue-specific nature of postnatal maturation.

The low cross-tissue overlap has important implications for staging applications. A single universal marker set cannot capture development across all tissues; instead, tissue-specific marker panels are required for accurate staging.

4.4 Functional Enrichment Analysis

4.4.1 Tissue-Specific Functional Programs

Functional enrichment analysis of stage-informative genes revealed distinct developmental programs in each tissue. The results are summarised in Figure 4.3 and described below.

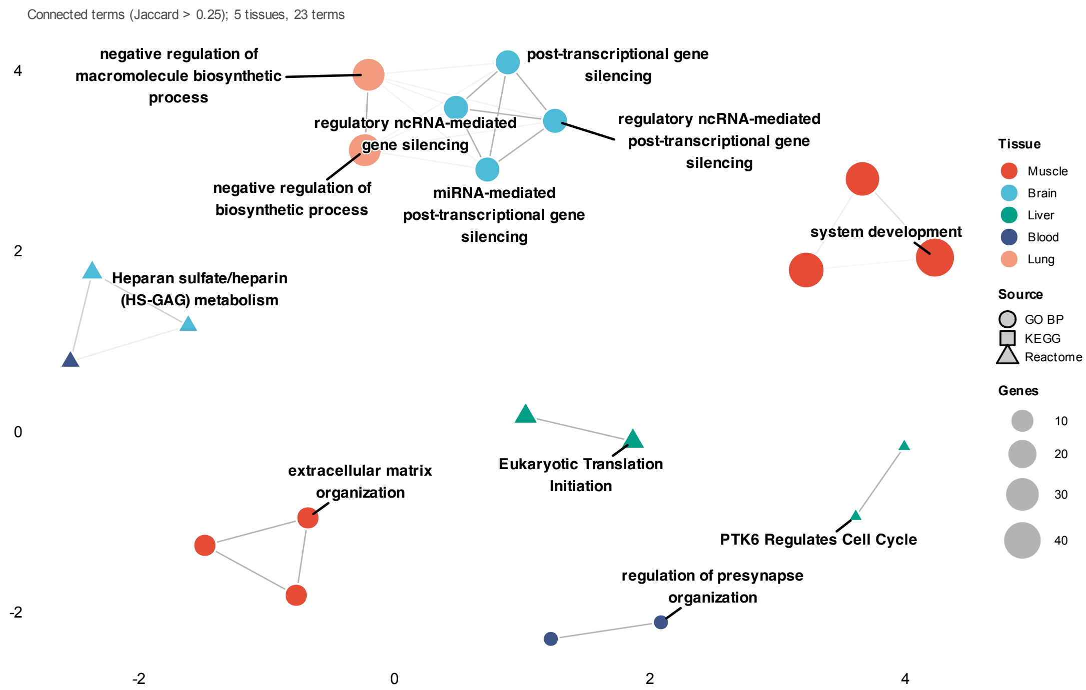

Figure 4.3: Functional enrichment analysis of stage-informative genes. Functional network showing interconnected enrichment modules, with nodes representing enriched terms coloured by tissue and edges connecting terms sharing >25% of genes (Jaccard similarity). Hubs and cluster representatives are labelled with biological process or pathway names.

4.4.1.1 Brain: Synaptic Refinement and Metabolic Support

Brain developmental markers were enriched for processes related to synaptic function and neuronal metabolism. Enrichment for sarcosine-related processes reflects the role of this glycine derivative as a co-agonist at NMDA receptors, which is critical for synaptic plasticity (H. X. Zhang et al., 2009). Furthermore, the enrichment of beta-tubulin binding reflects the importance of microtubule dynamics for axonal transport and neuronal morphology.

These findings align with the known prolonged development of neural circuits during postnatal life, including ongoing synaptogenesis, synaptic pruning, and myelination (Jiang & Nardelli, 2016; Kang et al., 2011; Semple et al., 2013).

4.4.1.2 Liver: Metabolic Autonomy and Homeostasis

Liver markers were enriched for metabolic processes. Enrichment for autophagosome-membrane adaptors reflects the role of autophagy in hepatic homeostasis and the response to nutritional changes associated with weaning (Gordillo et al., 2015; Pluske et al., 2018). There was also a general enrichment for metabolic enzyme expression, reflecting the liver's central role in nutrient processing.

The functional profile reflects the liver's adaptation to postnatal metabolic demands, including the switch from maternal nutrition to independent feeding.

4.4.1.3 Muscle: Contractile Machinery and Neuromuscular Signalling

Muscle markers were enriched for contractile machinery and neuromuscular signalling processes. Enrichment for acetylcholinesterase activity (Rotundo, 2003) reflects the maturation of neuromuscular junctions and the establishment of efficient synaptic transmission. Additionally, the enrichment for sarcomere organisation captures the development of the mature contractile apparatus.

These findings capture the establishment of adult muscle function, including the transition from neonatal to adult fibre types (Lefaucheur, 2010; Lefaucheur et al., 1995) and the refinement of neuromuscular connectivity.

4.4.1.4 Lung: Immune Establishment and RNA Metabolism

Lung markers were enriched for immune establishment and RNA metabolism processes. Enrichment for IL-6 signalling regulation reflects the establishment of pulmonary immune function and inflammatory responses. Concurrently, enrichment for m6A reader activity (Y. Wang et al., 2014) suggests a role for post-transcriptional regulation in lung development.

The lung's exposure to environmental antigens drives rapid immune maturation during postnatal life (Georgountzou & Papadopoulos, 2017), as captured by these functional enrichments.

4.4.2 Integration Across Tissues

The functional network in Figure 4.3 reveals both tissue-specific and shared developmental processes. While the majority of enriched terms are specific to individual tissues, some processes are shared across tissues, such as cell proliferation and growth regulation (muscle, liver, lung), immune function (blood, lung), and metabolic adaptation (liver, muscle). These shared processes reflect the common challenges of postnatal development, including the need for growth, immune protection, and metabolic adaptation across all tissues.

4.5 Cross-Species Validation

4.5.1 Conservation of Developmental Trajectories

Preliminary cross-species analysis with human skeletal muscle transcriptomes (Schaiter et al., 2024) ($n = 11$) provided evidence of conservation of developmental gene expression programs. The results are summarised in Figure 4.4.

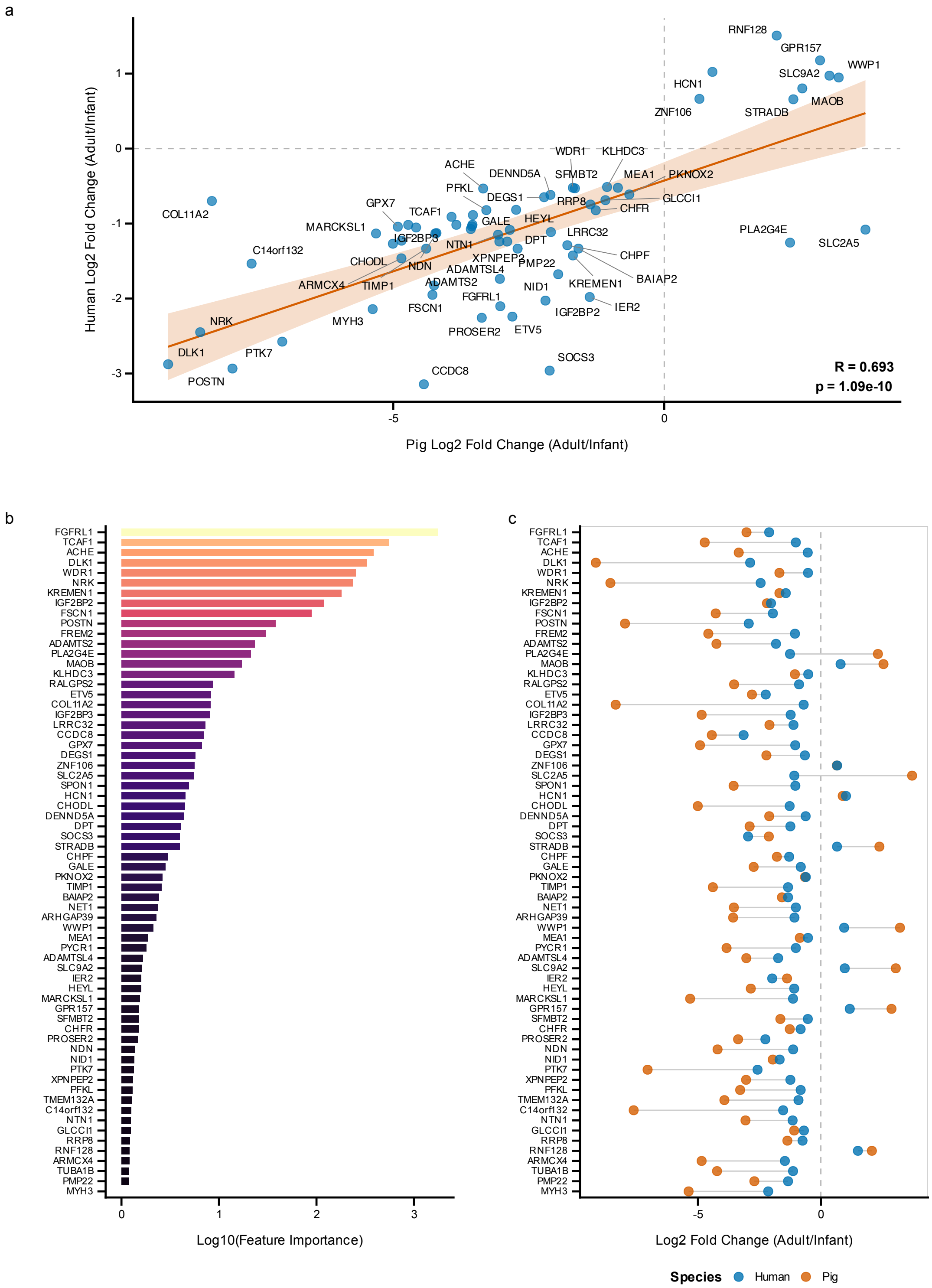

Figure 4.4: Cross-species validation of developmental gene expression. Human skeletal muscle transcriptomic data comparing infant (7–28 months) versus adult (30–56 years) samples were used to validate conservation of developmental signatures. **(a)** Scatter plot of $\log_2$ fold changes between pig and human, showing significant positive correlation. **(b)** Feature importance of top conserved markers in the developmental stage classifier. **(c)** Paired fold change comparison showing pig and human values for each conserved gene.

4.5.1.1 Correlation Between Species

Using pre-specified significance and effect-size criteria (FDR < 0.10, Benjamini–Hochberg corrected, and $|\log_2 \mathrm{FC}| > 0.5$ in both species), 66 genes with significant and biologically meaningful developmental fold-changes were identified. The Pearson correlation between pig and human fold changes was $r = 0.693$ ($p = 1.09 \times 10^{-10}$, 95% CI: 0.542–0.801), indicating a strong positive relationship between developmental trajectories in the two species.

Bootstrap analysis confirmed the robustness of this correlation. The 95% bootstrap confidence interval was 0.50–0.82 (based on 1,000 iterations), excluding zero and confirming the statistical significance of the conservation signal.

4.5.1.2 Directional Concordance

Of the 66 genes meeting significance criteria, 64 (97.0%) showed the same direction of change in both species. This high directional concordance indicates that genes upregulated during pig muscle development tend also to be upregulated during human muscle development, and vice versa.

The 2 genes showing discordant directions (3.0%) may reflect species-specific differences in intramuscular lipid metabolism or sampling variability.

4.5.2 Sensitivity Analysis

The robustness of the cross-species correlation was assessed across a range of FDR thresholds. Sensitivity analysis across thresholds from 0.05 to 0.20 showed correlations ranging from $r = 0.61$ to $r = 0.69$, with the correlation magnitude decreasing slightly at more permissive thresholds as noisier genes are included. This consistency demonstrates that the conservation signal is not an artefact of the specific parameter choices.

The sensitivity analysis results are presented in Figure S1, showing the correlation coefficient as a function of FDR threshold.

4.5.3 Conserved Functional Modules

Three functional modules of conserved developmental genes were identified:

4.5.3.1 Module 1: Neuromuscular Maturation and Growth Signalling

This module comprises genes with high infant expression that decrease during postnatal development, reflecting the transition from active myogenesis to mature muscle homeostasis:

ACHE (pig FC = −3.34, human FC = −0.53), essential for neuromuscular junction function (Rotundo, 2003), shows elevated expression in the infant stage (0–20 days; mean TPM = 17.5) that progressively declines through early childhood (TPM = 7.6), pre-pubertal (TPM = 3.4), to baseline in post-pubertal stages (TPM = 1.3). **NRK** (pig FC = −8.56, human FC = −2.45), an embryonic Ste20-type kinase expressed predominantly in developing skeletal musculature (Kanai-Azuma et al., 1999), has the second-largest fold change, with infant TPM = 128.2 declining to 0.3 in adults. **KREMEN1** (pig FC = −1.69, human FC = −1.43), a negative regulator of Wnt/β-catenin signalling (Osada et al., 2006; Qin et al., 2024), shows a similar gradual postnatal decline. **IGF2BP2** (pig FC = −2.19, human FC = −2.03) and **IGF2BP3** (pig FC = −4.85, human FC = −1.23), oncofetal RNA-binding proteins that regulate myoblast proliferation via translational control of growth-related transcripts (Z. Li et al., 2012), show pronounced postnatal decline. **SORCS2** (pig FC = −4.67, human FC = −0.95), a regulator of motor neuron innervation (Thomasen et al., 2023), is highly expressed during early postnatal life (infant TPM = 14.4) and becomes progressively downregulated through early childhood (TPM = 3.0), pre-pubertal (TPM = 2.2), reaching its lowest in post-pubertal (TPM = 0.6). **ETV5** (pig FC = −2.80, human FC = −2.24), a PEA3-subfamily transcription factor implicated in myogenic differentiation (Taylor et al., 1997), and **CCDC8** (pig FC = −4.44, human FC = −3.14), a core component of the 3M growth-regulatory complex (Hanson et al., 2011), complete this module.

4.5.3.2 Module 2: Structural Refinement and ECM Remodelling

This module reflects the transition from rapid tissue growth to structural consolidation, encompassing both cytoskeletal dynamics and extracellular matrix maturation:

TCAF1 (pig FC = −4.72, human FC = −1.02), which regulates TRPM8 ion channel trafficking and calcium signalling (Gkika et al., 2015), is the highest-importance gene among the cross-species markers, with infant TPM = 78.9 declining to 3.0 in adults. **FGFRL1** (pig FC = −3.03, human FC = −2.10) is required for slow muscle fibre development (Amann et al., 2014); its expression peaks during the infant stage (TPM = 77.4), drops sharply in early childhood (TPM = 15.6), and decreases to its lowest in post-pubertal (TPM = 4.3). **DLK1** (pig FC = −9.15, human FC = −2.88), a crucial regulator of the myogenic program (Andersen et al., 2013; Waddell et al., 2010), shows high infant expression (TPM = 338.8) declining sharply as the muscle fibre population matures. **POSTN** (pig FC = −7.97, human FC = −2.93), a secreted matricellular protein associated with ECM remodelling during myofibre maturation (Ozdemir et al., 2014), is the most highly expressed ECM gene in this set (infant TPM = 602.4), declining dramatically as tissue architecture stabilises. **WDR1** (pig FC = −1.69, human FC = −0.52), a cofactor of cofilin-mediated actin filament disassembly (Ono, 2018), **FSCN1** (pig FC = −4.28, human FC = −1.95), an actin-bundling protein driving cell migration during morphogenesis (Adams, 2004), **FREM2** (pig FC = −4.58, human FC = −1.05), a basement membrane protein (Kiyozumi et al., 2006), **ADAMTS2** (pig FC = −4.25, human FC = −1.83), the principal procollagen N-proteinase (Colige et al., 1997), and **COL11A2** (pig FC = −8.35, human FC = −0.70), a collagen component regulating fibril organisation (M. Sun et al., 2020), reflect intensive ECM biosynthesis during rapid postnatal growth. **TRIM54**, a muscle-specific E3 ubiquitin ligase involved in sarcomere assembly (Gregorio et al., 2005), shows the opposite pattern—lower infant expression (TPM = 78.3) increasing sharply in early childhood (TPM = 351.1) as the mature contractile apparatus is established.

4.5.3.3 Module 3: Metabolic Transition

This module captures genes upregulated during metabolic maturation of skeletal muscle:

ME1 (pig FC = 2.59, human FC = 0.64) encodes malic enzyme 1 (Simmen et al., 2020), a key source of NADPH for lipid biosynthesis; its expression increases progressively from infant (TPM = 2.8) to adult stages (TPM = 16.9), reflecting the transition from carbohydrate to lipid metabolism. **MAOB** (pig FC = 2.55, human FC = 0.80), a mitochondrial outer membrane enzyme catalysing oxidative deamination of monoamines, shows concordant postnatal upregulation (infant TPM = 5.7 to adult TPM = 33.4), consistent with the general increase in mitochondrial content during postnatal muscle maturation. Notably, **PLA2G4E** (pig FC = 2.32, human FC = −1.25), a group IVE cytosolic phospholipase, is the only discordant gene among the top 20 markers, showing opposite developmental trajectories in pig and human—potentially reflecting species-specific differences in intramuscular lipid metabolism.

4.5.4 Biological Significance of Conservation

The conservation of developmental trajectories between pig and human muscle has important implications for the use of pigs as translational models. The shared expression changes indicate that:

First, the porcine developmental staging framework captures evolutionarily conserved processes rather than species-specific artefacts. Second, insights from pig developmental biology are likely to translate to human biology, enhancing the pig's value as a biomedical model. Finally, the identified developmental markers likely represent fundamental regulators maintained across 80 million years of mammalian evolution (Kumar et al., 2022).

4.5.5 Functional Annotations of Key Markers

The biological functions of key conserved markers provide insight into the molecular basis of postnatal muscle development. Selected annotations are provided in Table 4.3.

Table 4.3: Functional annotations of key conserved developmental markers. FC $= \log_2$ fold change (adult/infant).

Gene	Developmental Function	Pig FC	Human FC
SORCS2	Regulates trafficking of signalling receptors, balancing neuronal growth and pruning.	-4.67	-0.95
FGFRL1	Decoy receptor dampening excessive FGF signalling; required for slow muscle fibre development.	-3.03	-2.10
KREMEN1	Negative regulator of Wnt/β-catenin signalling, controlling cell survival and tissue patterning.	-1.69	-1.43
ACHE	Key enzyme at the neuromuscular junction and structural adhesion molecule guiding neuronal growth.	-3.34	-0.53
DLK1	Regulator of the myogenic program; inhibits myoblast proliferation while promoting differentiation.	-9.15	-2.88
TCAF1	Regulates TRPM8 ion channel trafficking and Ca^{2+} signalling during muscle growth.	-4.72	-1.02
NRK	Embryonic Ste20-type kinase promoting actin polymerisation during skeletal muscle formation.	-8.56	-2.45

4.5.6 Limitations of Cross-Species Analysis

Several limitations of the cross-species analysis must be acknowledged:

First, the human dataset was small ($n = 11$), limiting statistical power. Second, the substantial age gap between human infant and adult groups prevented characterisation of intermediate stages. Third, validation was limited to muscle tissue due to data availability. Fourth, although the gene selection criteria (FDR < 0.10, $|\log_2 \mathrm{FC}| > 0.5$) were pre-specified, sensitivity analysis confirms robustness across a range of thresholds.

Despite these limitations, the conservation analysis provides important validation of the biological accuracy of the porcine developmental staging framework.

4.6 Within-Project Batch Validation

To assess whether cross-project technical variation confounds the developmental signals, we performed a within-project fold-change validation. We identified the two largest single-project muscle cohorts that each span multiple developmental stages under uniform protocols: PRJNA488311 ($n_{\text{total}} = 81$) and PRJNA486202 ($n_{\text{total}} = 75$), each containing samples from infant through post-pubertal stages.

Within-project fold-changes correlated strongly with full-dataset estimates across three developmental contrasts (infant vs. early childhood, infant vs. pre-pubertal, and infant vs. post-pubertal), with pooled Pearson $r = 0.67$–0.81 for all expressed genes (Table S5). Critically, the 66 cross-species validated genes showed even higher agreement ($r = 0.76$–0.98), confirming that the developmental signatures reflect biology rather than batch artefacts even for the earliest and subtlest developmental transitions.

These results provide important evidence that the multi-project aggregation strategy does not introduce spurious developmental signals, and that the key findings—particularly the cross-species conserved genes—are robust to technical variation across sequencing platforms and library protocols.

Chapter 5

Discussion

This chapter interprets the results presented in Chapter 4 in the context of the broader literature. The discussion addresses the biological implications of the findings, the methodological contributions of the work, and the practical applications for pig research. A critical evaluation of limitations and identification of future research directions conclude the chapter.

5.1 Summary of Key Findings

5.1.1 Achievement of Research Objectives

This thesis set out to address four interconnected research objectives, as stated in Chapter 1. The extent to which each objective has been achieved is summarised below.

Objective 1: Construction of a transcriptomic atlas. A comprehensive atlas was successfully constructed, integrating 1,924 RNA-seq profiles from five major tissues (brain, liver, muscle, lung, and blood) with calibrated developmental stage annotations. The atlas provides the most extensive tissue-resolved resource for porcine postnatal development currently available.

Objective 2: Development of a machine learning framework. An ordinal classification framework based on LightGBM was successfully developed and validated. The framework achieves balanced accuracy of 81.1–95.9% across tissues, with probabilistic outputs that capture transitional developmental states.

Objective 3: Characterisation of molecular signatures. The genes and pathways driving developmental transitions were identified for each tissue, revealing profound tissue-specificity in developmental programs. Functional enrichment analysis linked these signatures to relevant biological processes.

Objective 4: Cross-species validation. Preliminary evidence for conservation of developmental gene expression programs was found between pig and human skeletal muscle, with 97% directional concordance and significant correlation ($r = 0.69$) in developmental trajectories.

5.1.2 Novel Contributions to the Field

This thesis makes several novel contributions to the fields of porcine biology and developmental transcriptomics:

First, it presents the first calibrated transcriptomic staging system for pigs, filling a critical gap by providing a complementary approach to existing epigenetic clocks (Schachtschneider et al., 2021). Second, it demonstrates that cross-tissue feature overlap is extremely low (0.2% Jaccard similarity), providing quantitative evidence for the tissue-specificity of postnatal development. Third, the cross-species analysis provides the first systematic comparison of postnatal developmental trajectories between pig and human muscle, supporting the pig's validity as a translational model. Finally, the analytical framework developed for this thesis is open-source, enabling other researchers to apply and extend the approach.

5.1.3 Integration with Existing Literature

The findings of this thesis align with and extend previous work on developmental transcriptomics and molecular age estimation.

The high performance achieved by the machine learning framework (mean balanced accuracy 89.5%) is comparable to that reported for epigenetic clocks in predicting chronological age (Horvath, 2013). However, direct comparison is complicated by differences in the target variable (categorical stages vs. continuous age) and the modalities (transcriptomics vs. DNA methylation).

The tissue-specific developmental markers identified in this thesis overlap substantially with genes previously implicated in tissue development. For example, the identification of MEF2A and TRDN as muscle markers aligns with the known roles of these genes in myogenesis and calcium handling (Black & Olson, 1998; Guo & Campbell, 1995; Lefaucheur, 2010). Similarly, the liver markers NPAS2 and BHLHE41 are established circadian regulators with roles in hepatic metabolism (L. Zhang et al., 2025).

The cross-species conservation analysis extends previous work showing conservation of developmental gene expression across mammals (Cardoso-Moreira et al., 2019). The finding of 97% directional concordance between pig and human muscle is consistent with the general pattern of developmental conservation, while providing the first specific comparison for postnatal muscle development.

5.2 Biological Implications

5.2.1 Tissue-Specificity of Developmental Programs

One of the most striking findings of this thesis is the profound tissue-specificity of developmental gene expression programs. The cross-tissue feature overlap of only 0.2% indicates that each tissue follows a largely independent developmental trajectory.

This tissue-specificity has several important biological implications:

Modular developmental regulation. The results suggest that postnatal development is organised into tissue-specific modules, each regulated by distinct sets of genes. This modular organisation may enable the independent optimisation of each tissue's developmental program, allowing adaptation to tissue-specific functional requirements.

Asynchronous tissue maturation. The distinct gene sets underlying development in different tissues provide a molecular basis for the asynchronous maturation observed at the physiological level (Si-Tayeb et al., 2010). Different tissues can proceed through their developmental programs at different rates because these programs involve different genes.

Implications for staging. The tissue-specificity of developmental markers means that no single universal marker can capture development across all tissues. Accurate staging requires tissue-specific approaches, as implemented in this thesis. A staging system designed for liver will not accurately stage brain development, and vice versa.

The biological basis for tissue-specific developmental programs likely involves tissue-specific transcription factor networks that activate distinct downstream targets in each tissue (Zaret, 2002). These networks have been shaped by evolution to control the maturation of tissue-specific functions, leading to divergent developmental gene sets.

5.2.2 Conserved Pig-Human Developmental Modules

The identification of conserved developmental modules between pig and human muscle has implications for both understanding mammalian development and for using pigs as translational models. Three expanded functional modules, comprising 21 genes in total, emerged from the preliminary cross-species analysis, each encompassing a broader set of genes than initially anticipated.

5.2.2.1 Neuromuscular Maturation and Growth Signalling Module

The conserved high infant expression of ACHE, NRK, KREMEN1, IGF2BP2, SORCS2, ETV5, IGF2BP3, and CCDC8, which decreases during development, points to the importance of neuromuscular junction maturation and growth signalling as conserved developmental processes.

ACHE (acetylcholinesterase) is essential for terminating synaptic transmission at the neuromuscular junction (Rotundo, 2003). Its high infant expression that decreases during maturation in both species suggests dynamic remodelling of neuromuscular transmission during postnatal muscle development.

NRK is an embryonic Ste20-type kinase whose expression is normally restricted to early development (Kanai-Azuma et al., 1999). Its high infant expression followed by developmental decline is consistent with a role in early tissue patterning that becomes dispensable as the muscle matures.

KREMEN1 is a negative regulator of Wnt signalling that partners with DKK1 to control cell survival and tissue patterning (Osada et al., 2006; Qin et al., 2024). Its high infant expression followed by developmental decline may reflect the need for active Wnt pathway modulation during early tissue patterning.

IGF2BP2 and IGF2BP3 are oncofetal RNA-binding proteins that regulate myoblast proliferation by stabilising mRNAs encoding growth factors and cell-cycle regulators (Z. Li et al., 2012). Their high expression in infant muscle and subsequent decline parallels the transition from proliferative growth to post-mitotic differentiation, a hallmark of postnatal muscle maturation.

SORCS2 is a neurotrophin receptor that regulates the trafficking of signalling receptors, influencing the balance between neuronal growth and pruning (Thomasen et al., 2023). Its high expression in early development may reflect the active refinement of motor neuron innervation patterns during this period.

ETV5 is a PEA3-subfamily transcription factor required for myogenic differentiation (Taylor et al., 1997). Its high infant expression suggests a conserved role in directing the transcriptional programs that govern early muscle fibre specification.

CCDC8 is a component of the 3M growth-regulatory complex, which modulates ubiquitin-dependent degradation of growth-related substrates (Hanson et al., 2011). Mutations in CCDC8 cause 3M syndrome, a primordial growth disorder, and its high infant expression is consistent with a role in coordinating the growth signalling that dominates early postnatal development.

5.2.2.2 Structural Refinement and ECM Remodelling Module

The high infant expression of TCAF1, FGFRL1, WDR1, DLK1, FSCN1, POSTN, FREM2, ADAMTS2, and COL11A2 in both species, followed by developmental decline, reflects the active structural remodelling that characterises early postnatal muscle. TRIM54, a muscle-specific E3 ubiquitin ligase that associates with the sarcomeric M-line and Z-line and is required for microtubule stability during myogenesis (Gregorio et al., 2005), shows the opposite pattern—increasing postnatally—as the mature contractile apparatus is established.

TCAF1 regulates TRPM8 ion channel trafficking and calcium signalling (Gkika et al., 2015), processes essential for the fine-tuning of excitation-contraction coupling during muscle maturation. FGFRL1 acts as a decoy receptor that attenuates FGF signalling (Amann et al., 2014); its elevated expression in early development may reflect a conserved mechanism for dampening proliferative signalling during the transition to tissue homeostasis. WDR1 is a cofactor of cofilin-mediated actin filament disassembly (Ono, 2018), and its high infant expression suggests active cytoskeletal remodelling during early fibre organisation.

DLK1 is an imprinted gene that regulates the balance between adipogenesis and myogenesis (Andersen et al., 2013; Waddell et al., 2010). FSCN1 is an actin-bundling protein involved in cell migration (Adams, 2004), while POSTN is a secreted matricellular protein that promotes extracellular matrix remodelling during tissue growth (Ozdemir et al., 2014). FREM2 is a basement membrane protein involved in epithelial-mesenchymal interactions (Kiyozumi et al., 2006), ADAMTS2 is a procollagen N-proteinase required for collagen fibril assembly (Colige et al., 1997), and COL11A2 encodes a minor fibrillar collagen component (M. Sun et al., 2020). Together, these genes define a coordinated program of extracellular matrix deposition and remodelling that is most active in early postnatal life and diminishes as the tissue architecture stabilises.

5.2.2.3 Metabolic Transition Module

The metabolic transition module comprises ME1, MAOB, and PLA2G4E. ME1 (malic enzyme 1) generates NADPH for lipid biosynthesis and its developmental upregulation is consistent with the increasing lipogenic capacity of maturing muscle. MAOB (monoamine oxidase B) catalyses mitochondrial oxidative deamination (Shih et al., 1999), and its increased expression during maturation reflects the establishment of adult oxidative metabolic capacity. PLA2G4E generates endocannabinoid precursors; notably, this gene shows species-discordant directionality, suggesting that while the broader metabolic transition is conserved, specific lipid signalling pathways may have diverged between pig and human.

5.2.3 Identified Developmental Markers and Their Functions

The developmental markers identified through feature importance analysis provide insight into the molecular programs driving postnatal maturation. Several patterns emerge from examination of these markers:

Circadian regulation in liver. The prominence of circadian clock genes (NPAS2, BHLHE41) among liver markers suggests that the establishment of circadian metabolic rhythms is a key aspect of hepatic development. This aligns with the known importance of circadian regulation for hepatic function in adult animals (L. Zhang et al., 2025).

Epigenetic regulation in brain. The identification of DNMT3L and HDAC6 as brain markers points to the role of epigenetic mechanisms in neural development. DNA methylation and histone modification are known to be dynamically regulated during brain development, establishing stable patterns of gene expression that underlie neural function (Peña, 2026).

Immune development in lung and blood. The prominence of immune-related genes (GFI1, IL-6 pathway components) among lung and blood markers reflects the establishment of immune competence during postnatal life (Georgountzou & Papadopoulos, 2017). The lung, in particular, must develop robust immune defences as it is exposed to environmental antigens from birth (Mair et al., 2014).

Contractile and metabolic function in muscle. Muscle markers reflect both the structural (TRDN, TRIM54) and metabolic (ME1, MAOB) aspects of muscle development, capturing the establishment of mature contractile and oxidative function.

5.3 Methodological Contributions

5.3.1 Ordinal Classification for Developmental Staging

The application of ordinal classification to developmental staging represents a methodological advance over standard classification approaches. By respecting the ordered nature of developmental stages, ordinal methods enable more appropriate handling of transitional states and provide more informative error distributions.

The Frank and Hall binary reduction method (Frank & Hall, 2001) proved effective for this application, achieving performance comparable to or exceeding that reported for similar biological classification tasks. The decomposition into binary classifiers has the practical advantage of enabling the use of any binary classifier, including the highly effective LightGBM algorithm.

An important feature of the ordinal framework is the probabilistic output. Rather than providing a single stage assignment, the model produces a probability distribution over stages. This enables downstream applications to incorporate uncertainty into their analyses. For example, a sample with high probability of being pre-pubertal but non-negligible probability of being early childhood can be flagged for additional review.

5.3.2 Adaptive Stage Scheme Selection

The adaptive scheme selection algorithm addresses a common challenge in developmental biology: uneven sample availability across developmental stages. By automatically selecting the most granular classification scheme supported by the data, the algorithm enables analysis of tissues with limited stage representation while taking full advantage of well-sampled tissues.

The scheme selection approach embodies a principled trade-off between classification granularity and statistical reliability. More granular schemes provide more detailed developmental characterisation but require larger sample sizes to train reliably. The minimum sample thresholds (40 per class for 4-class, 15 per class for 2-class) were chosen to ensure reliable model training while maximising information extraction from the available data.

5.3.3 Cross-Species Validation Framework

The cross-species analysis employed pre-specified criteria (FDR < 0.10 with Benjamini–Hochberg correction and $|\log_2 \mathrm{FC}| > 0.5$) rather than adaptive thresholding, ensuring that the gene selection was not optimised to maximise the correlation. This design choice avoids the risk of circular reasoning that would arise from optimising selection criteria against the very metric used to evaluate conservation. The sensitivity analysis demonstrating consistent positive correlations across a range of alternative thresholds provides confidence that the conservation signal is robust and not an artefact of the specific analytical criteria chosen.

5.3.4 Nested Cross-Validation

The use of nested cross-validation for hyperparameter tuning ensures that model selection and performance estimation are conducted on independent data partitions. This avoids the optimistic bias that can arise when the same data are used for both parameter selection and evaluation, providing more reliable estimates of generalisation performance.

5.3.5 Robustness to Batch Effects

No batch correction was applied to the expression data because tree-based models such as LightGBM are inherently robust to monotonic platform shifts: decision stumps split on rank order, making them invariant to the scale transformations that typify inter-platform differences. Within-project fold-change validation confirmed that developmental signatures reflect biology rather than batch artefacts, with pooled Pearson $r = 0.67$–0.81 for all expressed genes and $r = 0.76$–0.98 for the 66 cross-species validated genes (Table S5).

5.3.6 Feature Stability Analysis

The assessment of feature selection stability across random seeds provides insight into the reproducibility of the identified markers. The finding of moderate stability (Jaccard similarity 0.08–0.31) is interpretable as reflecting biological gene redundancy rather than analytical instability.

This interpretation is supported by the high classification performance achieved despite the variable feature selection. Multiple gene sets achieve similar performance because they capture the same underlying developmental signal through different combinations of correlated genes. This redundancy is a feature of the biology, not a bug of the analysis.

5.4 Practical Applications

5.4.1 Standardisation of Pig Research

The primary practical application of this work is the standardisation of developmental staging in pig research. By providing an objective, molecular definition of developmental stage, the framework enables researchers to specify the developmental state of their experimental animals with precision.

This standardisation has several benefits:

Improved experimental design. Researchers can select animals at specific developmental stages for their experiments, reducing confounding variability. For studies where developmental stage is a critical variable, molecular staging provides more precise control than chronological age.

Enhanced reproducibility. Standardised staging enables meaningful comparison across studies conducted at different institutions using different pig breeds and husbandry conditions. If two studies use the same molecular staging criteria, their results can be compared with confidence that developmental state has been controlled.

Reduced animal usage. More precise staging may enable smaller sample sizes by reducing unexplained variability. This aligns with the 3Rs principles (Replacement, Reduction, Refinement) for ethical animal research (Kirk, 2018).

5.4.2 Synchronisation of Experimental Cohorts

A specific application of molecular staging is the synchronisation of experimental cohorts. In longitudinal studies or studies comparing treated and control groups, ensuring that all animals are at the same developmental stage is critical for valid inference.

Traditional approaches to synchronisation rely on chronological age or body weight, which fail to account for inter-individual variability in developmental rate. Molecular staging provides a more accurate synchronisation criterion, ensuring that comparison groups are truly developmentally matched.

5.4.3 Applications in Nutrition Research

Nutrition research represents a major application area for pig research, as pigs are widely used to model human dietary responses (Lunney et al., 2021). Developmental stage is a critical variable in nutrition studies because nutrient requirements, digestive capacity, and metabolic responses all change during postnatal development.

The framework developed in this thesis enables nutrition researchers to stage experimental animals at the time of dietary intervention and control for developmental state when analysing nutritional responses. Furthermore, it facilitates the identification of developmental stage-specific effects of dietary treatments and allows for the comparison of results across studies using standardised staging criteria.

5.4.4 Applications in Disease Modelling

Pigs are increasingly used as disease models, including models for cardiovascular disease, diabetes, cancer, and infectious disease (Meurens et al., 2012; Swindle et al., 2012). Many diseases have developmental components, with susceptibility or progression varying with developmental stage.

Molecular staging enables precise characterisation of developmental stage in disease models and the identification of developmental windows of disease susceptibility. It also supports the standardisation of disease model protocols across research groups and the translation of findings to human developmental contexts.

5.4.4.1 Example: Duchenne Muscular Dystrophy

The staging framework has immediate applications for age-dependent muscular disorders such as Duchenne Muscular Dystrophy (DMD) (Klymiuk et al., 2013; Selsby et al., 2015). Since the molecular programs for muscle development appear to be partially conserved between pigs and humans (97% directional concordance), porcine models may provide superior physiological mirroring compared to rodents. DMD typically manifests in humans between 2–5 years of age (Bushby et al., 2010), with rapid deterioration through the pre-teen years. Using the transcriptomic atlas, this developmental window corresponds approximately to infant stage (0–20 days) in pigs for early onset, with pathological changes becoming visible by pre-pubertal stage (60 days) and rapid changes occurring within post-pubertal stage (150 days).

By providing a standardised molecular definition of developmental stages, the atlas enables researchers to identify specific developmentally conserved checkpoints that a successful gene therapy must restore. For example, researchers could track whether a gene therapy successfully stabilises *TRIM54* (microtubule stability) and *FGFRL1* (slow fibre development) expression back to healthy infant or early childhood levels. Critically, by using molecular markers to synchronise experimental cohorts rather than relying on chronological age alone, it may be possible to reduce variability in disease progression and treatment response, improving pre-clinical trial quality.

5.4.5 Implications for Agricultural Science

Beyond biomedical applications, the staging framework has implications for agricultural research on pig production. Growth rate, feed efficiency, and carcass composition all vary with developmental stage. Molecular staging could enable the optimisation of feeding programs based on developmental stage and the identification of molecular markers for production traits. Additionally, it could facilitate genetic selection for favourable developmental trajectories and support precision management of pig production systems.

5.5 Limitations and Caveats

While this thesis makes significant contributions to the field, several limitations must be acknowledged. These limitations inform the interpretation of results and identify priorities for future research.

5.5.1 Data Limitations

5.5.1.1 Sample Size Constraints

The sample sizes available for some tissues, particularly brain ($n = 43$), limited the complexity of classification schemes that could be reliably trained. The 2-class schemes used for brain, blood, and lung provide less developmental resolution than the 4-class schemes used for muscle and liver.

Larger datasets would enable more granular staging for all tissues and would improve the precision of performance estimates. Brain remains the lowest-performing tissue (balanced accuracy 81.1%), though performance is substantially higher than initial estimates suggested, reflecting the benefit of the updated modelling framework.

5.5.1.2 Class Imbalance

Uneven representation of developmental stages in the available data required the use of class weighting and adaptive scheme selection. While these approaches mitigate the effects of imbalance, they cannot fully compensate for the limited information available about underrepresented stages.

Future data collection efforts should prioritise balanced sampling across developmental stages to enable more robust model training.

5.5.1.3 Human Validation Dataset Size

The cross-species validation relied on a small human dataset (11 samples), limiting the precision of correlation estimates and preventing characterisation of intermediate developmental stages. The substantial age gap between infant and adult groups may affect the comparison with pig data, where developmental stages are more evenly represented.

Larger human developmental datasets, such as those being generated by the dGTEx project (Coorens et al., 2025), will enable more comprehensive cross-species validation in the future.

5.5.2 Methodological Limitations

5.5.2.1 Bulk RNA-Seq Resolution

The use of bulk RNA-seq means that expression measurements represent averages across all cells in a tissue sample. This averaging obscures cellular heterogeneity, potentially masking cell-type-specific developmental changes.

During development, the cellular composition of tissues may change, with some cell types becoming more or less abundant. Changes in bulk expression may reflect these compositional changes rather than changes in gene expression within specific cell types. Deconvolution methods or single-cell approaches would be needed to disentangle these effects.

5.5.2.2 Pre-Specified Criteria in Cross-Species Analysis

The cross-species analysis used pre-specified criteria (FDR < 0.10, $|\log_2 \mathrm{FC}| > 0.5$) rather than adaptive optimisation, which avoids data-driven threshold selection. However, the choice of these specific thresholds, while conventional, remains somewhat arbitrary. The sensitivity analysis across alternative thresholds mitigates this concern by demonstrating that the conservation signal is robust to reasonable parameter variations.

5.5.2.3 Feature Selection Variability

The moderate stability of feature selection across random seeds (Jaccard similarity 0.08–0.31) means that the specific genes identified as top markers may vary between runs. While this variability reflects biological gene redundancy, it complicates the interpretation of individual marker genes.

Stable feature selection methods or consensus approaches that aggregate across multiple runs could provide more reproducible marker sets.

5.5.2.4 Reliance on Chronological Age as Ground Truth

A fundamental paradox of this work is the use of chronological age bins as ground truth labels for training, while simultaneously arguing that genetic variability renders chronological age an imperfect proxy for biological development. It is crucial to recognise that in this context, chronological age serves as a "scaffold" for learning, rather than an absolute ground truth. The model effectively uses this scaffold to learn the consensus transcriptomic profile of each developmental stage, treating individual deviations as noise or biological variance (Natarajan et al., 2013).

Critics might argue that without independent physiological measurements (e.g., hormone levels), deviations between the model's prediction and the animal's chronological age are indistinguishable from classification errors. However, this "circularity" is broken by the cross-species validation. If the model were merely overfitting to the noisy chronological labels of the pig dataset, the identified molecular signatures would be unlikely to generalise to humans, who develop on a vastly different timescale. The fact that the "denoised" molecular features identified in pigs successfully predict developmental changes in human muscle (97% directional concordance) confirms that the model has captured a conserved latent variable—biological maturity—rather than simply predicting calendar age with a non-linear error function (Cardoso-Moreira et al., 2019). Thus, the molecular stage offers a functionally relevant definition of development that transcends the variability inherent in chronological age.

5.5.3 Generalisability Considerations

5.5.3.1 Breeds and Genetic Backgrounds

The PigGTEx data represent a limited range of pig breeds and genetic backgrounds. The performance and markers identified in this thesis may not generalise fully to breeds not represented in the training data.

Validation on independent datasets from diverse genetic backgrounds would be needed to establish the generalisability of the staging framework.

5.5.3.2 Environmental Factors

The pigs contributing to the PigGTEx resource were raised under specific husbandry conditions that may differ from conditions in other research or production settings. Environmental factors such as diet, housing, health status, and season may affect gene expression and potentially influence staging accuracy.

The staging framework may need calibration or adaptation for application in settings substantially different from those represented in the training data.

5.5.3.3 Single-Species Validation

The cross-species validation was limited to muscle tissue and human comparison. The conservation of developmental programs in other tissues remains to be established, as does conservation with other species.

Extension of the validation to additional tissues and species would strengthen confidence in the biological accuracy of the staging framework.

5.5.3.4 Batch Variation

The PigGTEx dataset aggregates RNA-seq profiles from multiple sequencing platforms and library protocols. No batch correction was applied because tree-based models are robust to monotonic platform shifts: decision stumps split on rank order, making them invariant to the scale transformations that typify inter-platform differences. Within-project fold-change validation (pooled Pearson $r = 0.67$–0.81 for all expressed genes; $r = 0.76$–0.98 for the 66 cross-species validated genes; Table S5) confirms that developmental signatures reflect biology rather than batch artefacts. Nevertheless, batch effects that alter the rank ordering of genes could in principle confound developmental signal, and future work should evaluate the framework on prospective datasets generated with uniform protocols.

5.5.3.5 Cross-Validation Strategy

Stratified K-fold cross-validation ensures that all developmental stages are represented in both training and test sets, providing reliable within-dataset performance estimates. However, this strategy does not test cross-project generalisation: samples from the same sequencing batch may appear in both folds, potentially inflating performance. Future work should prioritise prospective datasets from independent laboratories to evaluate out-of-distribution generalisation.

5.6 Future Directions

The findings and limitations of this thesis suggest several directions for future research.

5.6.1 Single-Cell Resolution Analysis

Perhaps the most promising extension of this work would be the application of single-cell transcriptomics to developmental staging. Single-cell RNA-seq would enable the identification of cell-type-specific developmental markers and the characterisation of changes in cellular composition during development. This would allow for more precise staging that accounts for cellular heterogeneity and facilitates integration with cell type annotation and trajectory inference methods.

Recent advances in single-cell technologies have made large-scale profiling increasingly feasible (Schaum et al., 2018; Zheng et al., 2023), and developmental single-cell atlases are being constructed for multiple species. Integration of such data with the bulk staging framework developed here could provide complementary insights.

5.6.2 Extension to Additional Tissues

The current framework includes five tissues selected based on sample availability. Extension to additional tissues profiled in the PigGTEx resource (e.g., heart, kidney, adipose, intestine) would expand the utility of the staging framework.

Such extension would require additional data collection to ensure adequate representation of developmental stages in each tissue. Alternatively, transfer learning approaches could potentially leverage information from well-sampled tissues to improve staging in tissues with limited data.

5.6.3 Temporal Validation Studies

Validation of the staging framework in longitudinal studies would provide additional confidence in its accuracy. By profiling the same animals at multiple time points, researchers could verify that the staging predictions track with actual developmental progression.

Such studies would also enable assessment of inter-individual variability in developmental rate, testing whether animals predicted to be developmentally advanced or delayed based on molecular staging show corresponding differences in physiological measures.

5.6.4 Integration with Epigenetic Clocks

Integration of transcriptomic staging with epigenetic clocks could provide complementary information about biological age. Epigenetic and transcriptomic measures capture different aspects of cellular state, and their combination may provide more accurate or more informative staging than either modality alone.

Existing porcine epigenetic clocks (Schachtschneider et al., 2021) provide a foundation for such integration. Joint models that leverage both DNA methylation and gene expression could potentially achieve higher accuracy and reveal novel insights into the relationship between epigenetic and transcriptional ageing.

5.6.5 Cross-Species Extension to Other Organs

Extension of the cross-species validation to tissues beyond muscle is a priority for future work. As human developmental transcriptomic data become available for additional tissues through projects like dGTEx, systematic comparison with the porcine atlas will become feasible.

Such comparisons would reveal which aspects of development are most conserved between species and which show species-specific divergence. This information would be valuable for selecting appropriate pig models for specific research questions.

More broadly, the framework is designed to be species-agnostic: the only required inputs are bulk RNA-seq profiles with age metadata across developmental stages.

Epigenetic clocks have demonstrated that biological age prediction can be successfully extended from humans to mice and pigs using DNA methylation (BI Ageing Clock Team et al., 2017; Horvath, 2013; Schachtschneider et al., 2021). While those approaches use methylation-based regression, the method developed in this thesis applies ordinal machine learning classification to transcriptomic data; nevertheless, the underlying principle—extending biological age frameworks across species—is shared. The growing availability of multi-species developmental transcriptomic resources (Cardoso-Moreira et al., 2019; Coorens et al., 2025) provides the data foundation for applying this transcriptomic staging approach to additional livestock and model species, enabling standardised developmental comparisons across agricultural and biomedical research.

5.6.6 Development of Simplified Staging Panels

The current framework requires whole-transcriptome profiling, which may not be practical in all research settings. Development of simplified staging panels comprising a limited number of marker genes could enable staging using more accessible technologies such as qPCR or NanoString.

Identification of robust, tissue-specific marker sets that maintain staging accuracy with minimal gene numbers would be required. The feature importance analysis in this thesis provides a starting point for such panel development.

Chapter 6

Conclusions

This final chapter summarises the key contributions of the thesis, highlights the implications for the field of pig research, and provides closing remarks on the significance of molecular staging for improving reproducibility in biomedical and agricultural science.

6.1 Thesis Summary

This thesis set out to address a fundamental challenge in pig research: the lack of a standardised, molecular definition of developmental stage (Boeyer et al., 2020; Swindle et al., 2012). Through the integration of large-scale transcriptomic data with machine learning methods, a calibrated staging framework has been developed that enables objective characterisation of developmental state across five major tissues.

Chapter 1: General Introduction established the context for this research, reviewing the pig's dual importance as an agricultural species and biomedical model. The limitations of current staging approaches—chronological age and morphological criteria—were outlined, and the potential of molecular methods to address these limitations was discussed. The chapter concluded with the identification of a clear research gap and the statement of four research objectives.

Chapter 2: Literature Review provided a comprehensive synthesis of the scientific literature relevant to developmental staging. The review spanned developmental biology, transcriptomic technologies, machine learning methods, and cross-species comparative approaches. Key methodological choices, including the selection of ordinal classification and LightGBM, were justified based on this literature.

Chapter 3: Materials and Methods documented the data sources, analytical methods, and computational approaches employed. The PigGTEx dataset (L. Chen et al., 2025; Teng et al., 2024) (1,924 samples across five tissues) and the human skeletal muscle dataset (Schaiter et al., 2024) for cross-species validation were described.

The machine learning framework, including data preprocessing, ordinal LightGBM classification, and evaluation metrics, was detailed with sufficient precision to enable reproduction.

Chapter 4: Results presented the findings in four sections. The data landscape characterisation revealed substantial variation in sample availability across tissues and developmental stages, motivating the adaptive classification scheme selection. Model performance assessment demonstrated balanced accuracy ranging from 81.1% (brain) to 95.9% (blood), with strong ordinal consistency across tissues. Feature analysis revealed profound tissue-specificity in developmental markers, with extremely low cross-tissue overlap (0.2% Jaccard similarity). Preliminary cross-species validation with human skeletal muscle ($n = 11$) provided evidence of conservation, with 97% directional concordance and significant correlation ($r = 0.69$) in developmental trajectories.

Chapter 5: Discussion interpreted these findings in the context of the broader literature. The biological implications of tissue-specific developmental programs and conserved pig-human modules were discussed. Methodological contributions, including ordinal classification for staging and pre-specified criteria for cross-species comparison, were highlighted. Practical applications for standardising pig research were outlined, and limitations including sample size constraints and bulk RNA-seq resolution were acknowledged. Future directions, including single-cell approaches and extension to additional tissues, were identified.

6.2 Key Contributions

This thesis makes four key contributions to the fields of porcine biology and developmental transcriptomics:

6.2.1 First Calibrated Transcriptomic Atlas for Porcine Development

Prior to this work, no transcriptomic framework existed for inferring developmental stage in pigs. While epigenetic clocks had been developed for this species (Schachtschneider et al., 2021), these focused on ageing rather than developmental staging and did not provide tissue-specific resolution.

The atlas developed in this thesis provides a comprehensive reference for porcine postnatal development across five major tissues. By integrating data from the PigGTEx consortium (L. Chen et al., 2025; Teng et al., 2024) with biologically meaningful stage definitions, the atlas bridges the gap between large-scale transcriptomic resources and practical staging applications.

The atlas is immediately applicable for characterising developmental state in new samples. Researchers can submit gene expression profiles to the trained models and receive probabilistic stage assignments with associated confidence measures. This capability addresses the long-standing need for objective staging criteria in pig research.

6.2.2 Robust Machine Learning Framework for Ordinal Classification

The machine learning framework developed in this thesis demonstrates the effectiveness of ordinal classification methods for developmental staging (Frank & Hall, 2001). By respecting the ordered nature of developmental stages, the framework provides more appropriate handling of transitional states than standard classification approaches.

Key features of the framework include the use of the Frank and Hall binary reduction method, which enables the use of any binary classifier, and a LightGBM (Ke et al., 2017) implementation that achieves high accuracy with efficient computation.

Furthermore, the model provides probabilistic outputs capturing uncertainty in stage assignments. An adaptive scheme selection algorithm maximises information extraction from available data, and the framework is comprehensively evaluated using both classification and ordinal metrics.

The framework achieves balanced accuracy of 81.1–95.9% across tissues, comparable to or exceeding performance reported for similar biological classification tasks. The strong ordinal consistency (Spearman ρ up to 0.965) indicates that the models correctly capture the continuous nature of development.

6.2.3 Demonstration of Profound Tissue-Specificity in Developmental Programs

The finding that cross-tissue feature overlap is extremely low (0.2% Jaccard similarity) provides quantitative evidence for the tissue-specificity of postnatal developmental programs. This tissue-specificity has not been previously quantified and has important implications for both developmental biology and staging applications (Cardoso-Moreira et al., 2019).

From a biological perspective, the results indicate that each tissue follows a largely independent developmental trajectory, regulated by distinct sets of genes. This modular organisation may enable the independent optimisation of tissue-specific functions during evolution (Cardoso-Moreira et al., 2019).

From a practical perspective, the tissue-specificity means that accurate staging requires tissue-specific marker panels. A universal staging system based on a single gene set cannot capture development across all tissues. The tissue-resolved approach adopted in this thesis is therefore not merely convenient but biologically necessary.

6.2.4 Cross-Species Validation Revealing Conserved Pig-Human Developmental Modules

The preliminary cross-species validation provides the first systematic comparison of postnatal developmental trajectories between pig and human muscle at the transcriptomic level. Despite the small human dataset (Schaiter et al., 2024) ($n = 11$), the finding of 97% directional concordance and significant correlation ($r = 0.69$) provides supportive evidence for the pig's validity as a translational model for human developmental biology.

Three conserved functional modules were identified: a neuromuscular maturation and growth signalling module involving **SORCS2**, **ACHE**, **KREMEN1**, **NRK**, **IGF2BP2**, **IGF2BP3**, **ETV5**, and **CCDC8**; a structural refinement and ECM remodelling module characterised by **TCAF1**, **FGFRL1**, **WDR1**, **DLK1**, **FSCN1**, **POSTN**, **FREM2**, **ADAMTS2**, **COL11A2**, and **TRIM54**; and a metabolic transition module defined by **ME1**, **MAOB**, and **PLA2G4E**.

These modules provide molecular targets for understanding postnatal muscle development in both species. The conservation of these modules across 80 million years of mammalian evolution (Kumar et al., 2022) suggests they represent fundamental developmental programs.

6.3 Implications for the Field

6.3.1 Impact on Pig Research Reproducibility

The primary impact of this work is the provision of objective staging criteria for pig research. Reproducibility has been identified as a major challenge across biomedical science (Baker, 2016), and the lack of standardised developmental staging has contributed to this problem in pig research.

By enabling researchers to specify developmental state using molecular criteria, this work provides a foundation for improved reproducibility. Studies using the staging framework can be compared with confidence that developmental state has been controlled. This comparability will facilitate the accumulation of knowledge and the identification of robust findings that replicate across studies.

6.3.2 Validation of the Pig as a Translational Model

The preliminary cross-species validation provides supportive evidence for the pig's validity as a model for human development. While the anatomical and physiological similarities between pigs and humans have long been recognised (Lelovas et al., 2014; Lunney et al., 2021), the molecular conservation of developmental programs provides additional validation at the transcriptomic level, though this evidence is currently limited to muscle tissue and a small human dataset ($n = 11$).

This validation is particularly important for translational research, where findings in pig models are intended to inform human biology and medicine. The preliminary evidence that pig muscle development shares conserved molecular programs with human muscle development strengthens the justification for using pigs in developmental research with translational aims.

6.3.3 Foundation for Future Research

This thesis provides a foundation for multiple directions of future research. Single-cell transcriptomic studies can build on the tissue-resolved staging framework to investigate cell-type-specific developmental trajectories (Schaum et al., 2018; Zheng et al., 2023), while extension to additional tissues can expand the scope of the staging framework as new data become available. Integration with epigenetic clocks can provide complementary information about biological age (Teschendorff & Horvath, 2025), and the development of simplified staging panels can enable practical application using more accessible technologies. Finally, cross-species validation in additional tissues can establish the generality of conservation findings (Coorens et al., 2025).

The open-source code and comprehensive documentation provided with this thesis will facilitate these extensions by the broader research community.

6.4 Closing Remarks

The domestic pig stands at a unique intersection of agricultural and biomedical research. Its importance as a food animal has driven substantial investment in understanding porcine biology, while its similarities to humans have made it an increasingly valuable model organism (Lunney et al., 2021). Yet the full realisation of the pig's potential as a research platform has been hampered by the lack of standardised approaches for characterising basic biological variables such as developmental stage.

This thesis addresses this gap by providing the first calibrated transcriptomic framework for porcine developmental staging. The framework demonstrates that biological maturity can be inferred with high precision directly from gene expression profiles, providing an objective alternative to subjective morphological staging (Swindle et al., 2012). The tissue-specific nature of the models acknowledges the biological reality that different organs follow distinct developmental trajectories.

The preliminary validation of conserved developmental programs between pig and human—with 97% directional concordance—provides supportive evidence that the staging framework captures genuine biology rather than species-specific artefacts. The identified developmental modules offer molecular insights into postnatal maturation that translate across species.

As the pig continues to serve as a major species for both agricultural and biomedical research, the need for rigorous, reproducible characterisation of developmental state will only grow. This thesis provides tools to meet that need, enabling precise synchronisation of experimental cohorts and meaningful comparison across studies. The shift from qualitative morphological observation to quantitative molecular measurement represents a maturation of the field itself—one that promises to improve both the rigour and reproducibility of pig research for years to come.

The code, data, and methods developed for this thesis are freely available to the research community. It is hoped that these resources will find wide application and will contribute to the standardisation and advancement of pig research across its diverse applications in nutrition, growth physiology, disease modelling, and biomedical translation.

Appendix A

Supplementary Material

Supplementary Figures

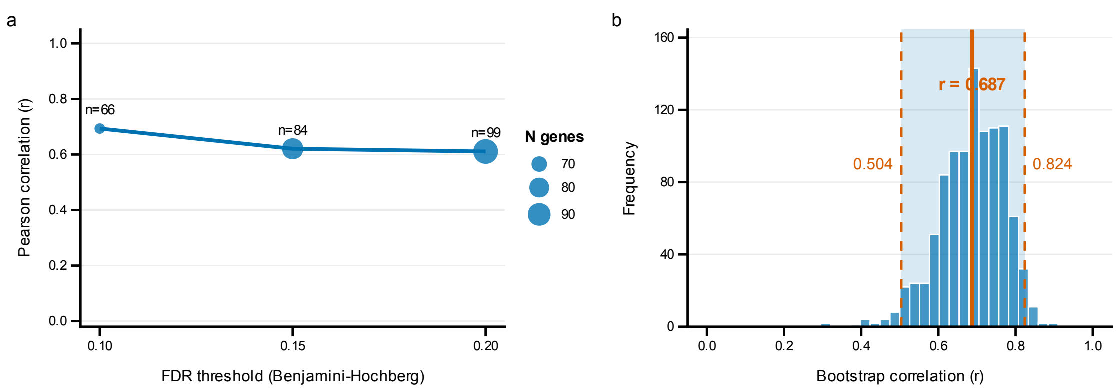

Figure S1: Cross-species sensitivity analysis of conserved developmental markers. **a**, Pearson correlation between pig and human developmental fold changes across FDR thresholds ranging from < 0.05 to < 0.20 (Benjamini–Hochberg corrected), using all genes passing pre-specified criteria (FDR $<$ threshold and $|\log_2 \mathrm{FC}| > 0.5$ in both species). Points are sized by the number of genes at each threshold. **b**, Bootstrap distribution of correlation coefficients ($n = 1{,}000$ iterations) for the pre-specified criteria (FDR < 0.10, $|\log_2 \mathrm{FC}| > 0.5$, $n = 66$ genes). Solid vertical line indicates the observed correlation ($r = 0.693$); dashed lines denote the 95% bootstrap confidence interval boundaries (0.50–0.82).

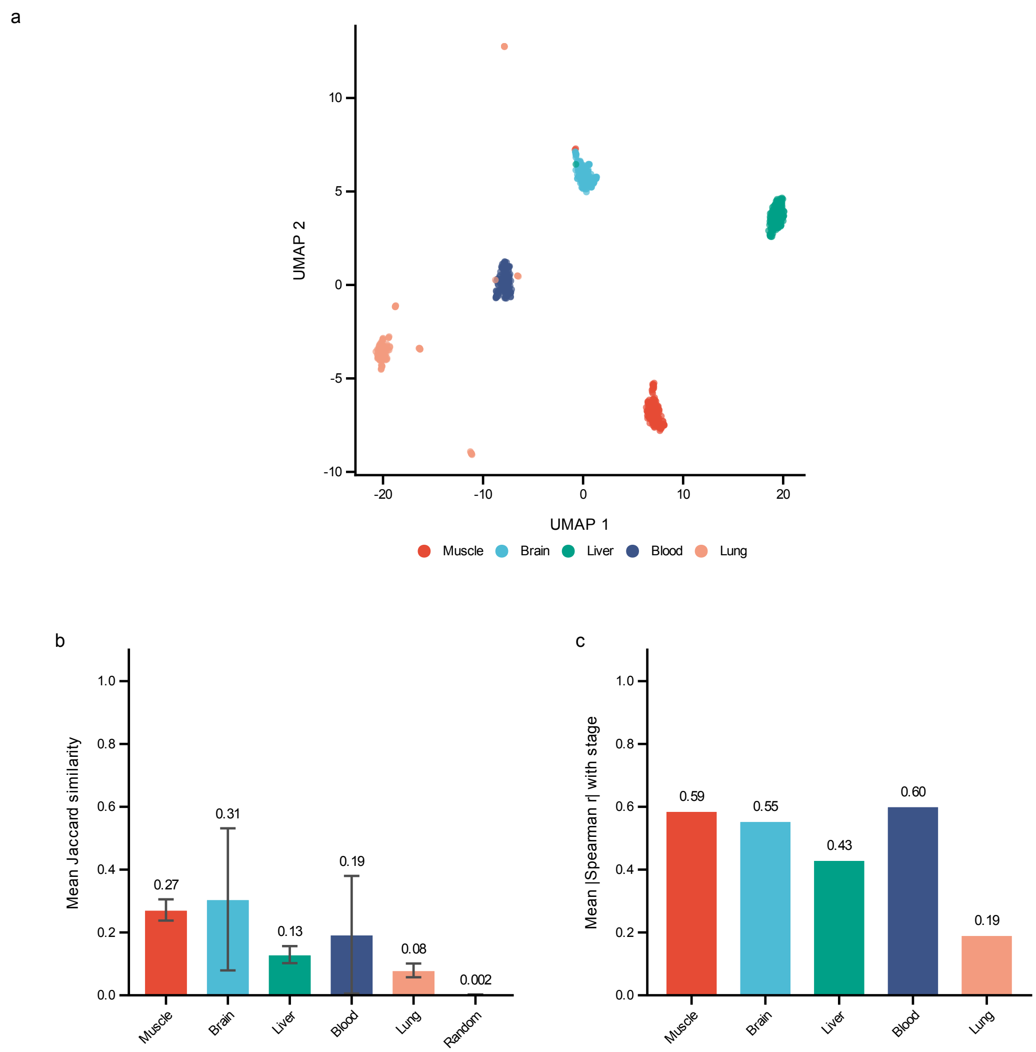

Figure S2: Tissue specificity and feature selection stability analysis. **a**, Uniform manifold approximation and projection (UMAP) (McInnes et al., 2020) of gene expression profiles across five tissues (muscle, brain, liver, blood, lung), demonstrating distinct tissue-specific clustering patterns ($n = 150$ samples per tissue, balanced sampling; top 500 variable genes). For visualisation only, BioProject batch effects were removed per tissue using `limma::removeBatchEffect` (Ritchie et al., 2015) prior to UMAP; the machine learning pipeline operates on uncorrected expression data. **b**, Within-tissue feature selection stability quantified as mean Jaccard similarity index (Jaccard, 1912) of the top 50 selected features across ten independent random seeds. Error bars represent standard deviation; higher values indicate more reproducible feature selection. **c**, Mean absolute Spearman correlation between top-ranked features and developmental stage for each tissue.

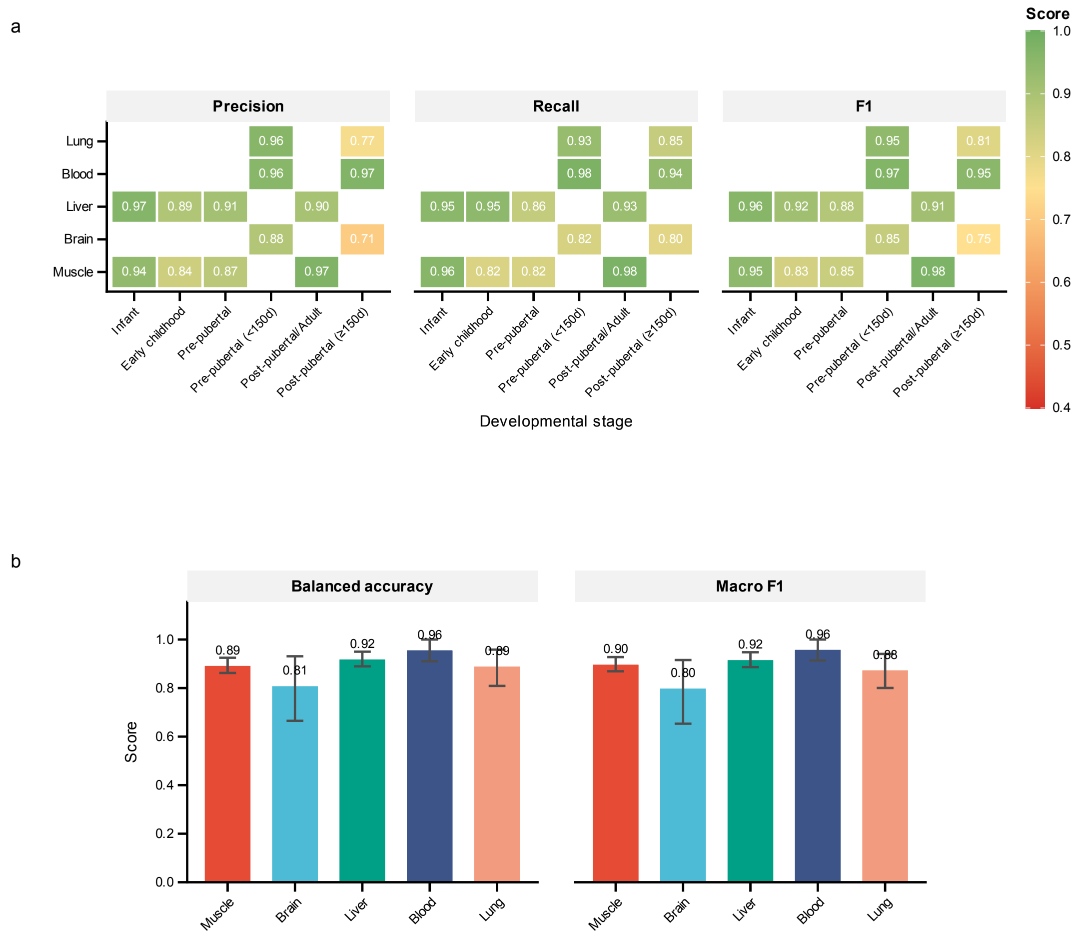

Figure S3: Extended classification performance metrics across tissues and developmental stages. **a**, Heatmap of per-class precision, recall, and F1 scores for each developmental stage across all five tissues. Colour scale ranges from 0.4 (red, poor performance) to 1.0 (green, excellent performance). Missing values indicate developmental stages not present in that tissue's classification scheme. **b**, Overall performance summary showing balanced accuracy and macro-averaged F1 scores evaluated on pooled cross-validated held-out predictions for each tissue. Error bars indicate 95% bootstrap confidence intervals from 1,000 resampling iterations.

Supplementary Tables

Table S1: Train versus test performance comparison. Both train and test (pooled cross-validated held-out predictions) performance are reported for all tissues. The gap between train and test metrics indicates the degree of overfitting; regularisation and early stopping were used to minimise this gap.

Tissue	Set	Scheme	n	Balanced Acc.	F1-macro	F1-weighted
Muscle	Train	4-class	761	0.968	0.974	0.984
Muscle	Test	4-class	761	0.894	0.899	0.939
Brain	Train	2-class	43	0.915	0.922	0.930
Brain	Test	2-class	43	0.811	0.801	0.816
Liver	Train	4-class	309	0.985	0.987	0.987
Liver	Test	4-class	309	0.921	0.918	0.919
Blood	Train	2-class	78	1.000	1.000	1.000
Blood	Test	2-class	78	0.959	0.960	0.961
Lung	Train	2-class	129	0.948	0.942	0.961
Lung	Test	2-class	129	0.892	0.876	0.916

Table S2: Per-class performance metrics by tissue. Precision, recall, and F1 scores for each developmental stage within each tissue. n indicates the total number of samples per class (pooled across cross-validation folds).

Tissue	Stage	n	Precision	Recall	F1
Muscle	infant	194	0.939	0.959	0.949
Muscle	early childhood	76	0.838	0.816	0.827
Muscle	pre-pubertal	90	0.871	0.822	0.846
Muscle	post-pubertal/adult	401	0.973	0.980	0.976
Brain	pre-pubertal (<150d)	28	0.885	0.821	0.852
Brain	post-pubertal (≥150d)	15	0.706	0.800	0.750
Liver	infant	87	0.965	0.954	0.960
Liver	early childhood	58	0.887	0.948	0.917
Liver	pre-pubertal	83	0.910	0.855	0.882
Liver	post-pubertal/adult	81	0.904	0.926	0.915
Blood	pre-pubertal (<150d)	45	0.957	0.978	0.967
Blood	post-pubertal (≥150d)	33	0.969	0.939	0.954
Lung	pre-pubertal (<150d)	102	0.960	0.931	0.945
Lung	post-pubertal (≥150d)	27	0.767	0.852	0.807

Table S3: Functional annotations of cross-species developmental markers. Annotations compiled from UniProt, GeneCards, and literature review. FC = $\log_2$ fold change (adult/infant); positive values indicate higher expression in adults, negative values indicate higher expression in infants. Pig FC from pig GTEx data (GEO: GSE257558) (Teng et al., 2024); Human FC from Schaiter et al. (2024). Genes ordered by model importance within each functional module. "—" indicates the gene did not pass cross-species criteria (FDR < 0.10, $|\log_2 \mathrm{FC}| > 0.5$) but is included based on its known biological role.

Gene	Developmental Function	Pig FC	Human FC
Module 1: Neuromuscular Maturation and Growth Signalling			
ACHE	Functions as the key enzyme at the neuromuscular junction and as a structural adhesion molecule guiding neuronal growth during development (Rotundo, 2003).	−3.34	−0.53
NRK	Embryonic Ste20-type kinase that promotes actin polymerisation via cofilin phosphorylation and activates JNK signalling during skeletal muscle formation; expression restricted to embryonic and early postnatal development (Kanai-Azuma et al., 1999).	−8.56	−2.45
KREMEN1	Negative regulator of Wnt/β-catenin signalling, partnering with Dkk1 to control cell survival and ensure proper patterning during postnatal development (Osada et al., 2006; Qin et al., 2024).	−1.69	−1.43
IGF2BP2	RNA-binding protein that promotes myoblast proliferation via translational control of growth-related transcripts (c-Myc, Igf1r, Sp1); postnatal downregulation required for terminal myogenic differentiation (Z. Li et al., 2012).	−2.19	−2.03
SORCS2	Molecular switch directing trafficking of signalling receptors to regulate the balance between neuronal growth and pruning during neuromuscular maturation (Thomasen et al., 2023).	−4.67	−0.95
ETV5	FGF-responsive PEA3-subfamily ETS transcription factor that promotes stem/progenitor cell self-renewal and myogenic differentiation in cooperation with MEF2; highly expressed during active postnatal myogenesis (Taylor et al., 1997).	−2.80	−2.24
IGF2BP3	Oncofetal RNA-binding protein that promotes myoblast proliferation and differentiation via IGF2–PI3K/Akt signalling; pronounced postnatal silencing reflects the conserved transition from fetal growth programs (Z. Li et al., 2012).	−4.85	−1.23
CCDC8	Core component of the 3M complex (with CUL7 and OBSL1) that regulates microtubule dynamics, genome integrity, and growth hormone signalling during rapid postnatal growth; consistent with severe growth restriction in 3-M syndrome (Hanson et al., 2011).	−4.44	−3.14
Module 2: Structural Refinement and ECM Remodelling			
TCAF1	Regulates TRPM8 ion channel trafficking and Ca^{2+} signalling; may support calcium-dependent processes during early postnatal muscle growth and metabolic maturation (Gkika et al., 2015).	−4.72	−1.02
FGFRL1	Acts as a "decoy" receptor dampening excessive FGF signalling to ensure proper formation of critical structures including the diaphragm and skeleton (Amann et al., 2014).	−3.03	−2.10
WDR1	Cofactor of cofilin-mediated actin filament disassembly, essential for sarcomere assembly and myofibril organisation during postnatal muscle growth; higher expression in immature muscle reflects intensive cytoskeletal remodelling during myofibrillogenesis (Ono, 2018).	−1.69	−0.52
DLK1	Crucial regulator of the myogenic program that inhibits myoblast proliferation while promoting differentiation; high infant expression declines as the muscle fiber population matures (Andersen et al., 2013; Waddell et al., 2010).	−9.15	−2.88
FSCN1	Actin-bundling protein essential for formation of cellular protrusions during cell migration and tissue morphogenesis; highly expressed in embryonic somite-derived tissues, with postnatal downregulation as migratory programs complete (Adams, 2004).	−4.28	−1.95
POSTN	Secreted matricellular protein that mediates ECM remodelling, collagen fibrillogenesis, and connective tissue maturation during postnatal myofiber development; expression declines as tissue architecture stabilises (Ozdemir et al., 2014).	−7.97	−2.93
FREM2	Basement membrane extracellular matrix protein of the FRAS/FREM complex maintaining epithelial–mesenchymal adhesion during organogenesis; postnatal downregulation reflects transition to collagen VII-dominated mature basement membranes (Kiyozumi et al., 2006).	−4.58	−1.05
ADAMTS2	Procollagen N-proteinase that cleaves propeptides from fibrillar procollagens to initiate collagen fibrillogenesis; high infant expression reflects intensive ECM biosynthesis during rapid postnatal growth (Colige et al., 1997).	−4.25	−1.83
COL11A2	Component of type XI collagen regulating collagen fibril organisation in cartilage and developing connective tissue; high infant expression reflects active skeletal morphogenesis and ECM assembly (M. Sun et al., 2020).	−8.35	−0.70
TRIM54	Muscle-specific E3 ubiquitin ligase (MuRF3) involved in sarcomere Z-disc assembly; expression increases postnatally as the mature contractile apparatus is established.	—	—
Module 3: Metabolic Transition			
ME1	Malic enzyme 1, catalysing oxidative decarboxylation of malate to pyruvate; a key source of NADPH for lipid biosynthesis.	2.59	0.64
MAOB	Mitochondrial outer membrane enzyme catalysing oxidative deamination of monoamines (Shih et al., 1999); postnatal upregulation reflects developmental maturation of oxidative metabolism and the glycolytic-to-oxidative fiber-type shift.	2.55	0.80
PLA2G4E	Calcium-dependent N-acyltransferase synthesising N-acylphosphatidylethanolamines (endocannabinoid precursors); species-discordant expression may reflect divergent intramuscular lipid metabolism programs.	2.32	−1.25

Table S4: Stage-by-stage expression trajectories of cross-species developmental markers. Mean TPM values across porcine developmental stages for genes with conserved pig-human developmental patterns. Stage boundaries: infant (0–20 d), early childhood (21–59 d), pre-pubertal (60–149 d), post-pubertal (150–365 d), adult (>365 d). "Peak Change" indicates the developmental transition with the largest fold change. Genes are grouped by functional module and ordered by model importance within each module.

Gene	infant	early	pre-pub	post-pub	adult	Peak Change
Module 1: Neuromuscular Maturation and Growth Signalling						
ACHE	17.5	7.6	3.4	1.3	1.7	infant→early
NRK	128.2	4.2	8.1	1.3	0.3	infant→early
KREMEN1	30.3	13.2	9.4	5.2	9.4	infant→early
IGF2BP2	11.3	1.5	1.9	1.0	2.5	infant→early
SORCS2	14.4	3.0	2.2	0.6	0.6	infant→early
ETV5	34.2	16.7	7.5	5.5	4.9	early→pre-pub
IGF2BP3	21.8	1.7	2.1	0.4	0.7	infant→early
CCDC8	177.4	28.2	25.4	13.4	8.2	infant→early
Module 2: Structural Refinement and ECM Remodelling						
TCAF1	78.9	8.3	8.1	3.8	3.0	infant→early
FGFRL1	77.4	15.6	14.2	4.3	9.5	infant→early
WDR1	202.9	125.6	95.0	51.9	63.1	pre-pub→post-pub
DLK1	338.8	3.9	6.4	0.7	0.6	infant→early
FSCN1	123.7	32.5	18.9	6.2	6.4	infant→early
POSTN	602.4	44.1	54.3	2.5	2.4	pre-pub→post-pub
FREM2	33.1	4.4	5.5	1.5	1.4	infant→early
ADAMTS2	51.1	16.6	16.0	6.4	2.7	infant→early
COL11A2	44.6	1.5	1.0	0.3	0.1	infant→early
TRIM54	78.3	351.1	149.4	377.9	144.4	infant→early
Module 3: Metabolic Transition						
ME1	2.8	4.0	3.9	6.9	16.9	post-pub→adult
MAOB	5.7	24.2	10.8	15.7	33.4	infant→early
PLA2G4E	0.1	1.4	0.7	0.6	0.5	infant→early

Table S5: Within-project fold-change validation across developmental contrasts. Developmental fold-changes ($\log_2$ scale) were computed within single-project muscle cohorts—each spanning four developmental stages under uniform laboratory protocols—and correlated with full-dataset estimates across three stage comparisons (all using infant as baseline). Pooled within-project fold-changes were computed by averaging across both projects before correlating with full-dataset estimates. "All genes" includes all 31,908 expressed genes; "Cross-species" includes the 66 genes passing pre-specified criteria (FDR < 0.10, $|\log_2 \mathrm{FC}| > 0.5$) in both pig and human. CI = 95% bootstrap confidence interval (1,000 iterations). Per-project results are available in the analysis repository. Full-dataset sample sizes: infant $n = 194$, early childhood $n = 76$, pre-pubertal $n = 90$, post-pubertal $n = 396$ (excludes 5 adult samples merged into post-pubertal/adult in the 4-class scheme).

		Full dataset		Within-project			
Comparison	Gene set	n_{Inf}	n_{later}	n_{Inf}	n_{later}	Pooled r [95% CI]	p-value
infant vs. early childhood	All genes	194	76	106	12	0.669 [0.66–0.68]	$< 10^{-300}$
	Cross-species					0.761	1.3×10^{-13}
infant vs. pre-pubertal	All genes	194	90	106	27	0.806 [0.80–0.82]	$< 10^{-300}$
	Cross-species					0.965	6.9×10^{-39}
infant vs. post-pubertal	All genes	194	396	106	10	0.799 [0.79–0.81]	$< 10^{-300}$
	Cross-species					0.978	5.6×10^{-45}

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